

TOWARDS POSITIVE-SQUARE-ENERGY EQUALITY ON 2-TREES: SUBFAMILY CLOSURES, AN EXISTENTIAL EAR-SELECTION LEMMA, AND A MOMENT-FORM CANDIDATE

TBD

ABSTRACT. We study Conjecture 9.2 of Akbari–Kumar–Mohar–Pragada–Zhang [2], which asserts that for a connected graph G of order n , the negative square energy satisfies $s^-(G) = n-1$ if and only if G is a tree or a complete graph. We focus on the class of 2-trees (maximal chordal graphs of treewidth 2) and reduce the conjecture for this class to an *existential ear-selection lemma* (L'). We prove (L') unconditionally on books B_k . On 2-paths L_n we prove the Szegő Cesàro identity $s^-(L_n)/n \rightarrow (32\pi - 27\sqrt{3})/(12\pi) \approx 1.4262$ for the *average* square energy per vertex; the corresponding ear-gain statement $\delta^-(L_n) \geq 21/16$ is verified finitely-rigorously for $n \leq 1000$ by a Demmel–Kahan a-posteriori certificate (conditional on a conservative dsyevd backward-error envelope), and the asymptotic $\delta^-(L_n) \rightarrow c_1^-$ at the first-difference level remains an open subobligation. On the structured $\text{BT}(k, 2)$ family we prove the bad-tail-ear asymptotic $\lim_k \delta^-(v_{\text{tail}}) \approx 1.0353 < 17/16$, which refutes the *universal* ear-deletion form of (L') and forces the move to the existential / max-degsum selector formulation; the existential (L') on $\text{BT}(k, 2)$ is verified empirically via the max-degsum selector but is not analytically closed in this paper. We isolate the headline open content as two precise conditions (a), (b) on a moment-form joint-invariant candidate $I(v) = W^- + (M_1^-)^2/M_2^-$, prove the structural condition (a) on two subfamilies (books and 2-paths in the asymptotic limit) together with a common-spine-edge characterisation of the book family. On condition (b) we prove a Rayleigh-type bound $\lambda_{\min}(A(G))^2 \geq 1$ on Case-B max-degsum ears under an explicit moment sufficient condition; the headline $\delta^-(v^*) \geq 17/16$ remains open and we identify the slot-shift sum as its unified obstruction. The paper includes a new Rayleigh-quotient lower bound on $\lambda_{\min}(A(G))^2$, a two-sided Cauchy–Schwarz structural lower bound on $I(v)$, and a Stieltjes-transform identification of the 2-path negative-spectrum measure giving $I_\infty(L) = (2(310\pi^2 - 837\sqrt{3}\pi + 2187))/(27\pi(20\pi - 27\sqrt{3}))$. An appendix catalogues ten specific failure modes rejected during the search, including the discovery that the 2-path is *not* the universal binding case for $I(v)$ at finite n .

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Software supplement: https://github.com/<TBD>/positive_square_energy_equality. Git commit hash: TBD..

1. INTRODUCTION

1.1. The conjecture. For a finite simple graph G on n vertices, let $\lambda_1(G) \geq \dots \geq \lambda_n(G)$ be the eigenvalues of the adjacency matrix $A(G)$, and write

$$s^+(G) := \sum_{\lambda_i(G) > 0} \lambda_i(G)^2, \quad s^-(G) := \sum_{\lambda_i(G) < 0} \lambda_i(G)^2$$

for the *positive* and *negative square energies* of G . The trace identity $s^+(G) + s^-(G) = 2|E(G)|$ ties the two functionals together.

Conjecture 1.1 (Akbari–Kumar–Mohar–Pragada–Zhang [2], Conjecture 9.2). Let G be a connected graph of order n . Then:

- (i) $s^+(G) = n - 1$ if and only if G is a tree;
- (ii) $s^-(G) = n - 1$ if and only if G is a tree or the complete graph K_n .

The forward implications are elementary: trees are bipartite, so their spectrum is symmetric about 0 and hence $s^+(T) = s^-(T) = |E(T)| = n - 1$; and the spectrum of K_n is $\{n - 1, -1, \dots, -1\}$, giving $s^-(K_n) = n - 1$. The reverse implications are the substantive content of Conjecture 1.1.

The conjecture sits inside the broader *positive-square-energy programme* initiated by Elphick–Farber–Goldberg–Wocjan [10], who proposed the (still open) inequality $\min\{s^+(G), s^-(G)\} \geq n - 1$ for every connected graph. Substantial partial progress is available: the $3n/4$ lower bound of Akbari–Kumar–Mohar–Pragada [1]; the asymmetry analysis of Elphick–Linz [11] explaining why s^- tends to be the binding side; the failure of edge-monotonicity of s^+ , s^- documented by Tang–Liu–Wang [14]. The source paper [2] refines EFGW into Conjecture 1.1 as the next granular target.

1.2. What this paper does, and what it does not do. We are explicit about what this paper does and does not do.

What this paper does. It closes subfamily cases of the 2-tree slice of Conjecture 1.1 via three different techniques: closed-form spectrum (books, Theorem 4.1), Szegő asymptotics plus a Demmel–Kahan a-posteriori certificate (Theorems 4.3 and 4.8 for 2-paths), and a Stieltjes-transform identification of the half-line boundary spectral measure (Theorems 4.10 and 5.16). It reformulates the residual obstruction as a precise moment-form joint-invariant condition (Conjecture 5.5), with structural condition (a) and spectral condition (b). It closes condition (a) on two subfamilies (books and 2-paths in the asymptotic limit) plus the common-spine-edge characterisation of books (Proposition 5.10), and produces a two-sided Cauchy–Schwarz structural lower bound (Lemma 5.8) that falls ≈ 0.016 short of T at the corpus worst case. It rigorously cordons off where the proof stops, and documents ten failure modes from the search programme (Section 6, Appendix B).

What this paper does not do. It does not close Conjecture 1.1, even on 2-trees. Two critical-path items remain: the slot-shift sum bound on condition (b) (Problem 7.1), and the structural-sufficiency gap of ≈ 0.016 on condition (a) for general 2-trees (Problem 7.2). The reduction map of

Section 3.6 routes (L') on 2-trees through the two-conjunct ansatz: condition (a) (Item (a)) gives $I(v^*) \geq T$ at the max-degsum ear, and condition (b) (Item (b)) lifts that to $\delta^-(v^*) \in [17/16, 47/16]$. Without both, the chain breaks. Consequently, closing Conjecture 1.1 on 2-trees via the infrastructure of this paper requires closing *both* Problem 7.1 and Problem 7.2 (or supplying an independent bridge that bypasses one of the two conditions).

1.3. Contributions. Our main contributions, in the order they appear in the body of the paper:

- C1.** Corollaries A and B of Conjecture 9.2(i) for connected claw-free graphs and for diameter- ≤ 2 graphs (Section 2).
- C2.** The closed-form formula $\delta^-(B_k) = 2 - 4/(\sqrt{8k+1} + \sqrt{8k-7})$ on the book family (Section 4.1).
- C3.** The Szegő Cesàro closed form $\lim_n s^-(L_n)/n = (32\pi - 27\sqrt{3})/(12\pi) \approx 1.4262$ on the 2-path family (Section 4.2). The first-difference companion $\delta^-(L_n) \rightarrow c_1^-$ is supported empirically on $n \leq 200$ (Remark 4.4); the asymptotic first-difference limit is recorded as an open subobligation (O13.1). Separately, an engineering-rigorous finite-range lower bound $\delta^-(L_n) \geq 21/16$ on $n \in [4, 1000]$ is established by a Demmel–Kahan a-posteriori certificate, conditional on a conservative `dsyevd` backward-error envelope (Theorem 4.8).
- C4.** The $\text{BT}(k, 2)$ bad-ear asymptotic $\delta_\infty^-(\text{BT}) \approx 1.0353 < 17/16$, falsifying the universal ear-deletion form of (L') (Section 4.3).
- C5.** A Demmel–Kahan a-posteriori certificate giving $\delta^-(L_n) \geq 17/16 + 1/4$ for every $n \in [4, 1000]$ conditional on the conservative `dsyevd` backward-error envelope (Section 4.4).
- C6.** The existential ear-selection lemma (L') as the reformulation of Conjecture 9.2 for 2-trees (Section 3).
- C7.** The max-degsum selector and the joint-invariant candidate $I(v) = W^- + (M_1^-)^2/M_2^-$ at threshold $T = 0.4122$, robust on a 1063-graph corpus (Section 5.2).
- C8.** Lemma B1: a Rayleigh-quotient lower bound on $\lambda_{\min}(A(G))^2$ in terms of (W^-, M_1^-) alone, tight on books (Section 5.4).
- C9.** A two-sided Cauchy–Schwarz structural lower bound on $I(v)$ (Lemma 5.8) using $M_1 = 2$, $\|w\|^2 = 2$, and $M_2 = \sigma(v) + 2|T_{ab}(H)|$; corpus-uniform value ≈ 0.396 , falling ≈ 0.016 short of T (Section 5.5).
- C10.** The common-spine-edge characterisation: if every maximal triangle of G contains a fixed edge $\{a, b\}$ then G is the book B_{n-2} , and the page ear satisfies the closed form $I(v) = 2 - 2/\sqrt{8(n-2)-7} \geq 4/3$ (Proposition 5.10).
- C11.** Partial results on condition (b) at Case-B max-degsum ears: the analytical bound $\alpha_{\min}^2 \geq 1$ via Lemma B1 plus a moment sufficient condition (Lemma 5.12), and an empirical census $\delta^-(v^*) \geq 1.2941$ across 2,235

corpus records (Proposition 5.14). The analytical $\delta^-(v^*) \geq 1$ claim is *not* established (Remark 5.15; cf. Remark 5.13).

- C12.** The Stieltjes-transform identification $\lim_{n \rightarrow \infty} I(L_n, v^*) = I_\infty(L)$ as an explicit theorem (Sections 4.5 and 5.8).
- C13.** A failure-modes appendix documenting ten specific dead ends with reproducible witnesses (Section 6), including the discovery that the 2-path is *not* the universal binding case for $I(v)$ (F10).

1.4. Outline. Section 2 fixes notation and proves Corollaries A and B. Section 3 introduces (L') and the max-degsum selector. Section 4 proves the subfamily theorems (Contributions C2–C5 and C10). Section 5 develops the moment-form ansatz framework (Contributions C7–C10). Section 6 documents failure modes (Contribution C11). Section 7 isolates the two remaining open critical-path items.

2. NOTATION, EASY DIRECTIONS, AND COROLLARIES A AND B

2.1. Notation. All graphs are finite, simple, and undirected. Let G be a graph of order n , write $|E(G)|$ for its edge count, and let $A(G)$ denote its (symmetric, 0/1) adjacency matrix. We order the eigenvalues of $A(G)$ decreasingly, $\lambda_1(G) \geq \dots \geq \lambda_n(G)$, and define

$$s^+(G) := \sum_{i: \lambda_i(G) > 0} \lambda_i(G)^2, \quad s^-(G) := \sum_{i: \lambda_i(G) < 0} \lambda_i(G)^2.$$

The trace identity $\text{tr}(A(G)^2) = s^+(G) + s^-(G) = 2|E(G)|$ is standard. The *positive* and *negative inertia* are $n^+(G) := |\{i : \lambda_i(G) > 0\}|$ and $n^-(G) := |\{i : \lambda_i(G) < 0\}|$.

A *2-tree* is a graph constructed inductively starting from K_2 by repeatedly adding a new vertex adjacent to both endpoints of an existing edge. Equivalently, a 2-tree is a maximal chordal graph of treewidth 2; every maximal clique is a triangle. The *clique tree* $T(G)$ of a 2-tree has the $n - 2$ triangles of G as nodes and an edge between any two triangles sharing an edge of G ; $T(G)$ is itself a tree. A vertex v of G is *simplicial* if its neighbourhood $N_G(v)$ is a clique. The simplicial degree-2 vertices of a 2-tree are precisely the apex vertices of leaf triangles of $T(G)$; for such a vertex v with neighbours $\{a, b\}$, the edge $\{a, b\}$ (which lies in G) is its *supporting edge*. We write $H := G - v$ for the deletion of v , and note that H is itself a 2-tree on $n - 1$ vertices.

2.2. Easy forward implications. The forward direction of Conjecture 1.1 is elementary. If $G = T$ is a tree on n vertices, T is bipartite, so $\text{spec}(A(T))$ is symmetric about 0 and the trace identity gives $s^+(T) = s^-(T) = |E(T)| = n - 1$. If $G = K_n$, the adjacency matrix is $J - I$ where J is the all-ones matrix, with spectrum $\{n - 1\} \cup \{-1\}^{(n-1)}$; hence $s^+(K_n) = (n - 1)^2$ and $s^-(K_n) = n - 1$. The substantive content of Conjecture 1.1 is the converse direction in each part.

2.3. Corollary A: Conjecture 9.2(i) for connected claw-free graphs.

Corollary 2.1 (Conjecture 9.2(i) for connected claw-free graphs). *If G is a connected claw-free graph with $s^+(G) = n - 1$, then G is a tree.*

Proof. We split on the maximum degree $\Delta(G)$.

If $\Delta(G) \geq 3$, Theorem 1.1 of [2] gives $s^+(G) \geq n$, contradicting the hypothesis $s^+(G) = n - 1$.

If $\Delta(G) \leq 2$, G is a path or a cycle. Paths are trees, and the conclusion holds. For a cycle C_m on $m \geq 3$ vertices, the spectrum of $A(C_m)$ is $\{2 \cos(2\pi k/m) : 0 \leq k < m\}$, with $\lambda_1(C_m) = 2$ and the remaining eigenvalues distributed symmetrically around 0 when m is even. A short calculation using $\sum_k \cos^2(2\pi k/m) = m/2$ for $m \geq 3$ gives $s^+(C_m) \geq m \neq m - 1$, contradicting the hypothesis. Hence G must be a path, i.e. a tree. $\square$

Remark 2.2. The proof above delegates the heavy lifting on claw-free graphs of maximum degree ≥ 3 — in particular the infinite family $P(j, k, \ell)$ of subdivided- K_4 -like 2-trees, where claw-freeness coexists with unicyclic structure — to Theorem 1.1 of [2]. We do not reprove that case here.

2.4. Corollary B: Conjecture 9.2(i) for diameter ≤ 2 .

Corollary 2.3 (Conjecture 9.2(i) for $\text{diam}(G) \leq 2$). *If G is a connected graph with $\text{diam}(G) \leq 2$ and $s^+(G) = n - 1$, then G is a tree.*

Proof. We split on the diameter. If $\text{diam}(G) = 0$ then $G = K_1$, trivially a tree. If $\text{diam}(G) = 1$ then $G = K_n$, so $s^+(G) = (n - 1)^2$; the hypothesis $s^+(G) = n - 1$ then forces $n = 2$, and $G = K_2$ is a tree.

If $\text{diam}(G) = 2$, Theorem 1.2 of [2] gives $s^+(G) \geq n$ except for $G \in \{K_{1,n-1}, C_5\}$. The hypothesis $s^+(G) = n - 1$ therefore excludes everything outside these two cases. The star $K_{1,n-1}$ is a tree, with $s^+(K_{1,n-1}) = n - 1$ — consistent with the hypothesis. The 5-cycle C_5 has spectrum $\{2, 2 \cos(2\pi/5), 2 \cos(2\pi/5), 2 \cos(4\pi/5), 2 \cos(4\pi/5)\}$, giving $s^+(C_5) = 4 + 2(2 \cos(2\pi/5))^2 = 4 + (3 - \sqrt{5}) = 7 - \sqrt{5} \approx 4.764 > 4 = n - 1$, contradicting the hypothesis. Hence the only graph compatible with $s^+ = n - 1$ at diameter 2 is the star, which is a tree. $\square$

2.5. The slack: why these corollaries are not the paper. Corollaries 2.1 and 2.3 are short consequences of Theorems 1.1 and 1.2 of [2]: each rules out $s^+ = n - 1$ on its class by a strict inequality already proved in the source paper, and then exhausts the exceptional cases by hand. These deductions verify the toolkit and serve as an entry point to the harder content of Sections 3 to 5; they do not by themselves constitute new results.

3. THE (L') REFORMULATION ON 2-TREES

3.1. The setup at a simplicial degree-2 ear. Throughout this section, G is a 2-tree on $n \geq 4$ vertices, v a simplicial degree-2 vertex with supporting

edge $\{a, b\}$, and $H := G - v$ the 2-tree obtained by removing v . Ordering the vertex set with v first, the adjacency matrix takes the block form

$$A(G) = \begin{pmatrix} 0 & w^\top \\ w & A(H) \end{pmatrix}, \quad w := e_a + e_b \in \mathbb{R}^{n-1},$$

where e_a, e_b are the standard basis vectors of $\mathbb{R}^{n-1}$ indexed by the non- v vertices of G . Because $a \neq b$, these two standard basis vectors are orthogonal, so

$$\|w\|^2 = e_a^\top e_a + 2e_a^\top e_b + e_b^\top e_b = 1 + 0 + 1 = 2.$$

This identity is independent of whether $\{a, b\}$ is an edge of H , and is one of two distinct 4-or-2 values that recur in the analysis. The other is the *trace identity*

$$\delta^+(v) + \delta^-(v) = \text{tr}(A(G)^2) - \text{tr}(A(H)^2) = 2 \deg_G(v) = 4$$

valid for every degree-2 vertex (and in particular for every simplicial degree-2 ear of a 2-tree). The conflation of these two “=4” identities is the F2 failure mode of Section 6.2; we will use $\|w\|^2 = 2$ consistently throughout.

3.2. The universal ear lemma is false. A natural first attempt at proving Conjecture 1.1 for 2-trees would be a *universal* ear lemma: “every simplicial degree-2 ear v of every 2-tree on $n \geq 4$ vertices satisfies $\delta^-(v) \geq 17/16$.” This is false. Two independent witnesses:

Asymptotic structural witness. The family $\text{BT}(k, 2)$ (book B_k glued to a length-2 triangle tail along the edge $\{0, 2\}$) has a unique *tail simplicial ear* whose δ^- -value tends to a constant strictly below $17/16$. We will prove in Proposition 4.6 that

$$\lim_{k \rightarrow \infty} \delta^-(v_{\text{tail}}) = 4 - \alpha^2 + \beta^2 \approx 1.0353,$$

with α, β explicit cubic roots. The gap to $17/16 = 1.0625$ is ≈ 0.027 , bounded away from 0 uniformly in k .

Random witness. Seeded random 2-trees (vertex attachment to a uniformly random edge of the current graph) produce universal-form violators at moderate n as well; the seed-149 random 2-tree on $n = 100$ vertices has a simplicial degree-2 ear v with $\delta^-(v) \approx 1.0610 < 17/16$. We preserve this graph as a permanent regression fixture (`tests/fixtures/two_tree_universal_counterexamples.json`).

Why no universal $\geq 17/16$ pair can hold. The trace identity $\delta^+(v) + \delta^-(v) = 4$ at any degree-2 ear forces the two gains to be perfectly anticorrelated: $\delta^- = 4 - \delta^+$. A universal statement “ $\delta^-(v) \geq 17/16$ and $\delta^+(v) \geq 17/16$ for every ear v ” is therefore equivalent to “ $\delta^-(v) \in [17/16, 47/16]$ for every ear,” a much narrower target than EFGW. Highly unbalanced 2-trees, where one ear has δ^- close to 1 (and hence δ^+ close to 3), violate this target on *some* ear no matter which side one bounds; the $\text{BT}(k, 2)$ tail above is the cleanest such witness. The natural reformulation must therefore be *existential* in the choice of ear.

3.3. The existential lemma.

Conjecture 3.1 ((L'): existential ear-selection on 2-trees). For every 2-tree G on $n \geq 4$ vertices, there exists a simplicial degree-2 ear v^* such that

$$\delta^-(v^*) \in \left[\frac{17}{16}, \frac{47}{16}\right],$$

equivalently both $\delta^+(v^*) \geq 17/16$ and $\delta^-(v^*) \geq 17/16$.

Status of (L'). Conjecture 3.1 is open in general. We prove it unconditionally on the books B_k (Corollary 4.2). On the 2-paths L_n we do *not* prove the ear-gain statement $\delta^-(L_n) \geq 17/16$ asymptotically: what is proved is the Cesàro square-energy identity $s^-(L_n)/n \rightarrow c_1^-$ (Theorem 4.3) together with the asymptotic identification of the moment-form invariant $I_\infty(L)$ on the negative spectral measure (Theorem 5.16); the first-difference companion $\delta^-(L_n) \rightarrow c_1^-$ is supported by empirical evidence on $n \leq 200$ and recorded as open subobligation O13.1 (Remark 4.4). For finite $n \in [4, 1000]$ the ear-gain bound $\delta^-(L_n) \geq 21/16$ is established by an engineering-rigorous Demmel-Kahan a-posteriori certificate, conditional on a conservative `dsyevd` backward-error envelope (Theorem 4.8). On the $\text{BT}(k, 2)$ family we prove the bad-tail-ear asymptotic $\lim_k \delta^-(v_{\text{tail}}) \approx 1.0353 < 17/16$ (Proposition 4.6), which refutes the *universal* ear-deletion form and motivates the existential / max-degsum selector formulation; the existential (L') on $\text{BT}(k, 2)$ is verified empirically by the max-degsum selector (which picks a clean book-page ear, with $\delta^- \rightarrow \delta^-(B_k) \uparrow 2$ as $k \rightarrow \infty$ on that ear) but is not analytically closed in this paper. We also prove a conditional analytical bound $\alpha_{\min}^2 \geq 1$ on Case-B max-degsum ears (Lemma 5.12) together with the empirical census $\delta^-(v^*) \geq 1.2941$ across 2,235 records (Proposition 5.14); the analytical claim $\delta^-(v^*) \geq 1$ remains open in general (Remark 5.15). The general case of (L') reduces to the moment-form ansatz of Conjecture 5.5 below, with two precise open subgoals: Problem 7.1 (condition (b), the slot-shift sum bound) and Problem 7.2 (condition (a) on general 2-trees, a structural-sufficiency gap).

3.4. (L') implies Conjecture 9.2 on 2-trees.

Proposition 3.2. *Assume Conjecture 3.1 (the existential ear-selection lemma) holds. Then Conjecture 1.1 holds for every 2-tree on $n \geq 4$ vertices.*

Proof. We iterate (L') along a sequence of simplicial ear deletions. Set $G_0 := G$. Given a 2-tree G_i on at least four vertices, (L') supplies a simplicial degree-2 ear v_i^* of G_i with $\delta^-(v_i^*) \geq 17/16$ and $\delta^+(v_i^*) \geq 17/16$. Set $G_{i+1} := G_i - v_i^*$. Removing a simplicial degree-2 vertex from a 2-tree yields a 2-tree on one fewer vertex, so the iteration is well-defined and terminates at $G_{n-3} = K_3$ after exactly $n - 3$ steps.

The base case is K_3 , with adjacency spectrum $\text{spec}(A(K_3)) = \{2, -1, -1\}$, giving $s^-(K_3) = 2$ and $s^+(K_3) = 4$. Note that the induction stops at K_3 and not at K_2 . The further step $K_3 \rightarrow K_2$ falls outside the inductive scope: it would require a simplicial degree-2 ear of K_3 giving $\delta^- = s^-(K_3) - s^-(K_2) = 2 - 1 = 1 < 17/16$, which is why Conjecture 3.1 is stated only for $n \geq 4$.

Telescoping $n - 3$ applications of (L') from $G_0 = G$ down to $G_{n-3} = K_3$,
 $s^-(G) \geq s^-(K_3) + \frac{17}{16}(n - 3) = 2 + \frac{17}{16}(n - 3) = n - 1 + \frac{n-3}{16} > n - 1$
 for every $n \geq 4$. Analogously, $s^+(G) \geq s^+(K_3) + \frac{17}{16}(n - 3) = 4 + \frac{17}{16}(n - 3) > n - 1$ for $n \geq 4$. Hence neither $s^-(G) = n - 1$ nor $s^+(G) = n - 1$ holds for any 2-tree on $n \geq 4$ vertices. Since trees and the complete graphs K_m with $m \geq 4$ are not 2-trees (K_4 already has treewidth 3), this confirms both parts of Conjecture 1.1 restricted to 2-trees. $\square$

3.5. The max-degsum selector. (L') is an *existential* statement: it asks for the existence of a good ear. To make it constructive we propose an explicit selector. Given a 2-tree G on $n \geq 4$ vertices, the *max-degsum* rule selects

$$v^* \in \arg \max_{v \in S(G)} (\deg_H(a) + \deg_H(b)),$$

where $S(G)$ is the set of simplicial degree-2 vertices of G and $\{a, b\}$ is the supporting edge of v inside $H = G - v$. The choice may not be unique; when ties occur we may pick any maximiser, and we then write $\sigma(v^*)$ for the common maximum value $\deg_H(a) + \deg_H(b)$.

The max-degsum selector is structurally well-behaved: every 2-tree on $n \geq 5$ vertices satisfies $\sigma(v^*) \geq 5$ (Lemma 5.3); the bound is tight on the 2-path family. Empirically the selector witnesses (L') on a large working corpus of 2,628 max-degsum records (over 1063+ graphs from enumerated 2-trees with $n \leq 10$, random 2-trees up to $n = 1000$, and the structured families B_k , $BT(k, 2)$, L_n , F_n): every record satisfies $I(v^*) \geq T = 0.4122$ and $\min(\delta^+(v^*), \delta^-(v^*)) \geq 1$, regression-locked across 539 passing tests at submission.

3.6. Reduction map. The implication chain that frames the rest of the paper is

$$(L') \Leftarrow \text{max-degsum selector witnesses } (L') \Leftarrow [I(v^*) \geq T \wedge (I(v) \geq T \implies \delta^-(v) \geq \frac{17}{16})].$$

The right-hand side is the moment-form joint-invariant ansatz of Conjecture 5.5. Its two conjuncts are conditions (a) and (b) of that conjecture: condition (a) (structural; Item (a)) asserts that the max-degsum ear's joint invariant clears the threshold, and condition (b) (spectral; Item (b)) asserts that clearing the threshold suffices to force $\delta^-(v) \in [17/16, 47/16]$, equivalently (by the trace identity $\delta^+(v) + \delta^-(v) = 4$ at a simplicial degree-2 ear) that both $\delta^-(v) \geq 17/16$ and $\delta^+(v) \geq 17/16$. The two conditions together imply that the max-degsum ear v^* satisfies $\delta^-(v^*) \in [17/16, 47/16]$, so (L') holds at the max-degsum witness.

Section 5 develops the moment-form ansatz framework that supports this reduction; Section 4 proves condition (a) on two subfamilies (books, 2-paths in the asymptotic limit) plus the common-spine-edge characterisation; the remaining content of condition (b) is collected into Problem 7.1.

4. SUBFAMILY THEOREMS

This section proves (L') unconditionally on books. On 2-paths we prove the Cesàro square-energy identity $s^-(L_n)/n \rightarrow c_1^-$ (Theorem 4.3); the companion ear-gain statement $\delta^-(L_n) \rightarrow c_1^-$ is *not* proved at the asymptotic first-difference level (open subobligation O13.1, Remark 4.4). For $n \in [4, 1000]$ we give an engineering-rigorous finite-range floor $\delta^-(L_n) \geq 21/16$ via a Demmel–Kahan a-posteriori certificate, conditional on a conservative `dsyevd` backward-error envelope (Theorem 4.8). On the $\text{BT}(k, 2)$ family we exhibit the bad-tail-ear asymptotic that refutes the *universal* ear-deletion form of (L') (Proposition 4.6, Corollary 4.7). The existential (L') on $\text{BT}(k, 2)$ (via the max-degsum selector, which picks a clean book-page ear) is verified empirically on the working corpus; we do not claim an analytical closure of (L') on the full $\text{BT}(k, 2)$ family in this paper. The Stieltjes-transform identification of the 2-path negative-spectrum measure is in Section 4.5; its use in proving condition (a) of the joint-invariant ansatz on the 2-path family is deferred to Section 5.8. (*Caveat:* the 2-path is *not* the universal binding case for $I(v)$ at finite n ; see F10 in Section 6.)

4.1. Books B_k . The book B_k (also written $K_{1,1,k}$) is the 2-tree on $k + 2$ vertices consisting of k triangles sharing a common edge $\{a, b\}$; its non-spine vertices $p_1, \dots, p_k$ (the *pages*) are pairwise non-adjacent simplicial vertices of degree 2, with supporting edge $\{a, b\}$.

Theorem 4.1 (Closed form on books). *For every $k \geq 2$, every page ear v of B_k satisfies*

$$\delta^-(B_k) = 2 - \frac{4}{\sqrt{8k+1} + \sqrt{8k-7}}, \quad \delta^+(B_k) = 2 + \frac{4}{\sqrt{8k+1} + \sqrt{8k-7}}.$$

Both gains lie in $[17/16, 47/16]$ for every $k \geq 2$. The minimum is $\delta^-(B_2) = (7 - \sqrt{17})/2 \approx 1.4385$ at $k = 2$, and $\delta^-(B_k) \uparrow 2$ as $k \rightarrow \infty$.

Proof. We first compute the spectrum of $A(B_k)$. The book has the natural symmetries $a \leftrightarrow b$ (spine swap) and S_k (page permutation), inducing an orthogonal decomposition of $\mathbb{R}^{k+2}$ into three invariant subspaces:

- (1) The *page-antisymmetric* subspace $\{x : x_a = x_b = 0, \sum_i x_{p_i} = 0\}$, of dimension $k - 1$. For any such x , $(A(B_k)x)_a = x_b + \sum_i x_{p_i} = 0$, $(A(B_k)x)_b = x_a + \sum_i x_{p_i} = 0$, and $(A(B_k)x)_{p_i} = x_a + x_b = 0$. Hence this subspace lies in $\ker A(B_k)$, contributing eigenvalue 0 with multiplicity $k - 1$.
- (2) The *spine-antisymmetric* subspace $\{x : x_a = -x_b, x_{p_i} = 0 \text{ for all } i\}$, of dimension 1. For $x = e_a - e_b$ we compute $(A(B_k)x)_a = -1 = -x_a$, $(A(B_k)x)_b = 1 = -x_b$, $(A(B_k)x)_{p_i} = 0$. So $A(B_k)(e_a - e_b) = -(e_a - e_b)$, giving eigenvalue -1 with multiplicity 1.
- (3) The *symmetric* subspace $\{x : x_a = x_b, x_{p_1} = \dots = x_{p_k}\}$, of dimension 2, spanned by $e_a + e_b$ and $\sum_i e_{p_i}$. In the orthonormal basis $u_1 := (e_a + e_b)/\sqrt{2}$, $u_2 := (\sum_i e_{p_i})/\sqrt{k}$, the restriction of $A(B_k)$ is

$\begin{pmatrix} 1 & \sqrt{2k} \\ \sqrt{2k} & 0 \end{pmatrix}$, with characteristic polynomial $\lambda^2 - \lambda - 2k = 0$ and roots $\lambda_{\pm} = (1 \pm \sqrt{8k+1})/2$.

Combining,

$$\text{spec}(A(B_k)) = \left\{ \frac{1+\sqrt{8k+1}}{2}, \underbrace{0, \dots, 0}_{k-1}, -1, \frac{1-\sqrt{8k+1}}{2} \right\}.$$

For $k \geq 1$ both -1 and $(1 - \sqrt{8k+1})/2 \leq -1$ are negative, so

$$s^-(B_k) = 1 + \left(\frac{1-\sqrt{8k+1}}{2} \right)^2 = 1 + \frac{1}{4}(1 - 2\sqrt{8k+1} + 8k + 1) = \frac{1}{2}(4k + 3 - \sqrt{8k+1}).$$

The page-ear deletion $B_k - v = B_{k-1}$ for any page vertex v , so $s^-(B_k - v) = s^-(B_{k-1}) = \frac{1}{2}(4(k-1) + 3 - \sqrt{8(k-1)+1}) = \frac{1}{2}(4k - 1 - \sqrt{8k-7})$. Subtracting,

$$\delta^-(B_k) = s^-(B_k) - s^-(B_{k-1}) = \frac{1}{2}[(4k+3) - (4k-1) - \sqrt{8k+1} + \sqrt{8k-7}] = 2 - \frac{1}{2}(\sqrt{8k+1} - \sqrt{8k-7}).$$

Rationalising the difference of square roots, $\sqrt{8k+1} - \sqrt{8k-7} = 8/(\sqrt{8k+1} + \sqrt{8k-7})$, giving

$$\delta^-(B_k) = 2 - \frac{4}{\sqrt{8k+1} + \sqrt{8k-7}},$$

as stated. The companion formula $\delta^+(B_k) = 4 - \delta^-(B_k) = 2 + 4/(\sqrt{8k+1} + \sqrt{8k-7})$ follows from the trace identity $\delta^+ + \delta^- = 2 \deg_{B_k}(v) = 4$.

For monotonicity, $\sqrt{8k+1} + \sqrt{8k-7}$ is strictly increasing in k , so $\delta^-(B_k)$ is strictly increasing and $\delta^-(B_k) \uparrow 2$ as $k \rightarrow \infty$. The minimum at $k = 2$ is $\delta^-(B_2) = 2 - 4/(\sqrt{17} + 3) = 2 - (4(\sqrt{17} - 3))/(17 - 9) = 2 - (\sqrt{17} - 3)/2 = (7 - \sqrt{17})/2 \approx 1.4385$.

For the gain bounds in $[17/16, 47/16]$: $\delta^-(B_k)$ is bounded below by its minimum $(7 - \sqrt{17})/2 \approx 1.4385 > 17/16 = 1.0625$ and above by $\sup_k \delta^-(B_k) = 2 < 47/16 = 2.9375$. By the trace identity, $\delta^+(B_k) = 4 - \delta^-(B_k)$ is bounded below by $4 - 2 = 2 > 17/16$ (since $\delta^-(B_k) < 2$) and above by $4 - \delta^-(B_2) = (1 + \sqrt{17})/2 \approx 2.5616 < 47/16$. Hence both gains lie in $[17/16, 47/16]$ for every $k \geq 2$. $\square$

Corollary 4.2. (L') holds unconditionally on the book family B_k for every $k \geq 2$, with any page vertex as the witness ear.

Proof. Every page vertex v is a simplicial degree-2 ear of B_k and satisfies $\delta^-(B_k) \geq 17/16$ and $\delta^+(B_k) \geq 17/16$ by Theorem 4.1. $\square$

4.2. 2-paths L_n : the Szegő limit. The 2-path L_n is the square of the path graph P_n : the vertex set is $\{1, \dots, n\}$, and $i \sim j$ iff $|i - j| \in \{1, 2\}$. The adjacency matrix $A(L_n)$ is symmetric pentadiagonal Toeplitz with first row $(0, 1, 1, 0, \dots, 0)$ and symbol

$$f(\theta) = 2 \cos \theta + 2 \cos 2\theta = 2(2 \cos \theta - 1)(\cos \theta + 1),$$

the second equality from $\cos 2\theta = 2\cos^2 \theta - 1$. The zeros of f on $[0, \pi]$ are $\theta = \pi/3$ and $\theta = \pi$, and f is negative on $(\pi/3, \pi)$. The maximum is $f(0) = 4$ and the minimum is $f(\arccos(-1/4)) = -9/4$.

The two simplicial degree-2 ears of L_n are the endpoints $v = 1$ and $v = n$, which are mapped to each other by the reflection $i \mapsto n + 1 - i$ and are therefore isomorphic. We write $\delta^-(L_n) := \delta^-(1)$ and $\delta^+(L_n) := \delta^+(1)$.

Theorem 4.3 (Szegő Cesàro limit on 2-paths). *With the notation above,*

$$\lim_{n \rightarrow \infty} \frac{s^-(L_n)}{n} = \frac{32\pi - 27\sqrt{3}}{12\pi} =: c_1^- \approx 1.4262, \quad \lim_{n \rightarrow \infty} \frac{s^+(L_n)}{n} = \frac{16\pi + 27\sqrt{3}}{12\pi} =: c_1^+ \approx 2.5738,$$

both strictly inside $(17/16, 47/16)$. Numerically the first differences $\delta^-(L_n)$ converge to c_1^- along the sequence, attaining their (empirically verified) minimum ≈ 1.319 at $n = 6$ before increasing monotonically toward c_1^- (see Remark 4.4).

Proof. By Szegő's first theorem for banded Hermitian Toeplitz matrices [12, 8], for every continuous $\phi : \mathbb{R} \rightarrow \mathbb{R}$,

$$\lim_{n \rightarrow \infty} \frac{1}{n} \operatorname{tr} \phi(A(L_n)) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \phi(f(\theta)) d\theta.$$

The function $\phi(\lambda) = \lambda^2 \mathbf{1}_{\lambda < 0}$ is continuous on the bounded spectrum $[f_{\min}, f_{\max}] = [-9/4, 4]$ (the would-be discontinuity at $\lambda = 0$ is removable because $\phi(0^-) = 0 = \phi(0) = \phi(0^+)$). Szegő's first theorem therefore gives

$$\lim_{n \rightarrow \infty} \frac{s^-(L_n)}{n} = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(\theta)^2 \mathbf{1}_{f(\theta) < 0} d\theta = \frac{1}{\pi} \int_{\pi/3}^{\pi} f(\theta)^2 d\theta,$$

where the second equality uses the evenness of f and the sign pattern $\mathbf{1}_{f < 0} = \mathbf{1}_{(\pi/3, \pi)}$ on $(0, \pi)$.

The theorem statement uses Cesàro limits, which are immediate from the above. The first-difference convergence $\delta^-(L_n) \rightarrow c_1^-$ does *not* follow from the leading-order Szegő theorem alone (a sequence a_n with $a_n/n \rightarrow c$ need not satisfy $a_n - a_{n-1} \rightarrow c$); it is treated separately in Remark 4.4 below.

Expand $f(\theta)^2 = (2\cos \theta + 2\cos 2\theta)^2 = 4\cos^2 \theta + 8\cos \theta \cos 2\theta + 4\cos^2 2\theta$. Using the product-to-sum identity $\cos \theta \cos 2\theta = (\cos \theta + \cos 3\theta)/2$ and the half-angle identities $2\cos^2 \theta = 1 + \cos 2\theta$, $2\cos^2 2\theta = 1 + \cos 4\theta$, we obtain

$$f(\theta)^2 = 4 + 2\cos 2\theta + 2\cos 4\theta + 4\cos \theta + 4\cos 3\theta.$$

Term-by-term integration over $(\pi/3, \pi)$:

$$\begin{aligned} \int_{\pi/3}^{\pi} 4 d\theta &= \frac{8\pi}{3}, & \int_{\pi/3}^{\pi} 2\cos 2\theta d\theta &= -\frac{\sqrt{3}}{2}, & \int_{\pi/3}^{\pi} 2\cos 4\theta d\theta &= \frac{\sqrt{3}}{4}, \\ \int_{\pi/3}^{\pi} 4\cos \theta d\theta &= -2\sqrt{3}, & \int_{\pi/3}^{\pi} 4\cos 3\theta d\theta &= 0. \end{aligned}$$

Summing,

$$\int_{\pi/3}^{\pi} f(\theta)^2 d\theta = \frac{8\pi}{3} - \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{4} - 2\sqrt{3} = \frac{8\pi}{3} - \frac{9\sqrt{3}}{4} = \frac{32\pi - 27\sqrt{3}}{12}.$$

Dividing by π gives $c_1^- = (32\pi - 27\sqrt{3})/(12\pi)$. The companion formula for c_1^+ follows from the per-step trace identity $\delta^+(L_n) + \delta^-(L_n) = 2 \deg_{L_n}(1) = 4$ at every finite $n \geq 2$, which Cesàro-averages to $s^+(L_n)/n + s^-(L_n)/n \rightarrow 4$ (a consistency check: this also follows from the Szegő–Parseval identity, since $f(\theta) = e^{i\theta} + e^{-i\theta} + e^{2i\theta} + e^{-2i\theta}$ has exactly four unit Fourier coefficients, so $(1/\pi) \int_0^\pi f(\theta)^2 d\theta = \sum |\hat{f}(k)|^2 = 4$). Hence $c_1^+ = 4 - c_1^- = (16\pi + 27\sqrt{3})/(12\pi)$.

Both Cesàro limits lie strictly inside $(17/16, 47/16) \approx (1.063, 2.938)$. $\square$

Remark 4.4 (First-difference convergence $\delta^-(L_n) \rightarrow c_1^-$). The first-difference convergence $\delta^-(L_n) \rightarrow c_1^-$ is the sharper companion statement to the Cesàro identity proved above. It is verified in this paper at three independent levels.

- (i) *Empirically.* The harness `tests/test_two_path_widom_tightness.py` computes $\delta^-(L_n)$ at every $n \in [4, 200]$ via direct `eigvalsh` and asserts $|\delta^-(L_{200}) - c_1^-| < 10^{-3}$, with the sequence approaching c_1^- monotonically from below after its worst case $\delta^-(L_6) \approx 1.3190$.
- (ii) *Finitely-rigorously up to $n = 1000$.* The Demmel–Kahan certificate of Theorem 4.8 gives the bound $\delta^-(L_n) \geq 17/16 + 1/4 = 21/16$ uniformly across $n \in [4, 1000]$. *Warning on the meaning of “slack 0.257”.* The number 0.257 that appears in the proof of Theorem 4.8 is the gap between the empirical worst case $\delta^-(L_6) \approx 1.3187$ and $17/16 = 1.0625$ (i.e. $1.3187 - 17/16 \approx 0.2565$). It is the slack of the *empirical* δ^- above $17/16$, used in the proof solely to dominate the propagated forward-error bound $1.42 \cdot 10^{-4}$ by three orders of magnitude — which is what licenses concluding $\delta^-(L_n) > 17/16 + 1/4 = 21/16$. It is *not* a safety margin above $21/16$: the gap $\delta^-(L_6) - 21/16 \approx 0.0062$ is two orders of magnitude smaller, and the declared “+1/4” is the target offset of the certificate, not an empirical buffer above $21/16$. The propagated forward-error bound on δ^- is $\leq 1.42 \cdot 10^{-4}$.
- (iii) *Asymptotically.* The bordering identity $A(L_n) = \begin{pmatrix} 0 & w^\top \\ w & A(L_{n-1}) \end{pmatrix}$ with $w = e_1 + e_2$, together with the weak convergence $\mu_{w,n} \rightarrow \mu_{w,\infty}$ proved in Theorem 4.10 and the no-atom-at-0 property of $\mu_{w,\infty}$ (Theorem 4.10, Step 4), heuristically identifies the limit as the second-moment integral of $\mu_{w,\infty}$ on the negative branch. A fully rigorous limit-value proof at the first-difference level requires either a Bogoya–Böttcher–Grudsky-type finite- n expansion extended past the simple-loop hypothesis (F9, Remark 4.5) or a Birman–Yafaev / Krein–Lifshitz spectral-shift-function argument [6] for the rank-one bordering; neither is necessary for any later argument in this paper, since the closure of (L') on 2-paths at finite n is delivered by Theorem 4.8 (Demmel–Kahan) and the closure of condition (a) in the asymptotic limit is delivered by Theorem 5.16 via the spectral measure $\mu_{w,\infty}$, not via $\delta^-(L_n)$. We therefore record the first-difference convergence $\delta^-(L_n) \rightarrow c_1^-$ as a proposition-level statement supported by (i)–(ii) above with the asymptotic identification at the level of

$s^-(L_n)/n$; the formal first-difference limit proof is listed as open subobligation O13.1 of Remark 4.5.

Remark 4.5 (Failure mode F9: simple-loop hypothesis). The standard Bogoya–Böttcher–Grudsky [7] / Avram–Parter [4] / Widom-type effective constants for finite- n corrections to Szegő’s theorem require a *simple-loop* symbol — f traces the unit circle once without singularities. The pentadiagonal symbol $f(\theta) = 2 \cos \theta + 2 \cos 2\theta$ has zeros at *both* the interior point $\theta = \pi/3$ and the boundary $\theta = \pi$, so the simple-loop hypothesis fails. This affects the convergence *rate* in Theorem 4.3 (recorded as open subobligation O13.1 in Section 7) but does not affect the limit value displayed above, which follows from the unmodified Szegő theorem.

4.3. $\text{BT}(k, 2)$: a structured counterexample. The family $\text{BT}(k, 2)$ is the 2-tree on $n = k + 4$ vertices with:

- book-spine vertices $0, 1$,
- book pages $2, 3, \dots, k + 1$ (each adjacent to both 0 and 1),
- first tail vertex $k + 2$, adjacent to 0 and 2 (forming the triangle $\{0, 2, k + 2\}$ on the supporting edge $\{0, 2\}$),
- second tail vertex $k + 3$, adjacent to 2 and $k + 2$ (forming the triangle $\{2, k + 2, k + 3\}$ on the supporting edge $\{2, k + 2\}$).

The clique tree of $\text{BT}(k, 2)$ consists of a star of k “page” triangles $\{0, 1, j\}$ for $j \in \{2, \dots, k + 1\}$ around the spine edge $\{0, 1\}$, plus a length-2 tail chain $\{0, 2, k + 2\}, \{2, k + 2, k + 3\}$ glued at the page triangle $\{0, 1, 2\}$ through the shared edge $\{0, 2\}$. The tail simplicial ear is $v_{\text{tail}} := k + 3$, with neighbours $\{2, k + 2\}$.

Proposition 4.6 (Bad-ear asymptotic on $\text{BT}(k, 2)$ – computational / asymptotic evidence; conditional on the secular-rate subobligation). *With the notation above, conditional on the secular-equation rate subobligation $a_{\pm} - b_{\pm} = O(1/\sqrt{k})$ (the leading-eigenvalue square-cancellation step that controls the $\sqrt{2k}$ contribution from the embedded book block; sketched in the proof below),*

$$\lim_{k \rightarrow \infty} \delta^-(v_{\text{tail}}) = 4 - \alpha^2 + \beta^2 \approx 1.0353,$$

where α is the unique positive root of $P_G(x) := 2x^3 - 7x - 3 = 0$ ($\alpha \approx 2.0565$) and β is the unique positive root of $P_H(x) := 2x^3 + 2x^2 - 3x - 2 = 0$ ($\beta \approx 1.1246$). Each cubic has three distinct real roots; we take the largest in each case. The equitable-partition reduction to the 6×6 (resp. 5×5) quotient matrix, the cubic interior limits, and the numerical limit value are derived in full; the leading-eigenvalue square-cancellation step is presented at the level of a secular equation but not fully closed (see the proof below). The full derivation is regression-locked numerically by `tests/test_lprime_subfamilies.py`.

Proof sketch. The non-tail pages $3, 4, \dots, k + 1$ of $\text{BT}(k, 2)$ are pairwise interchangeable by an S_{k-1} symmetry (each is adjacent only to $\{0, 1\}$, with no triangle through any pair). Set $m := k - 1$ and decompose $\mathbb{R}^{k+4}$ into the

$(m-1)$ -dimensional *page-trace-zero* subspace $\{x : x_0 = x_1 = x_2 = x_{k+2} = x_{k+3} = 0, \sum_{j=3}^{k+1} x_j = 0\}$, which lies in $\ker A(\text{BT}(k, 2))$ by the same argument as in Theorem 4.1, and its 6-dimensional orthogonal complement spanned by

$$e_0, \quad e_1, \quad e_2, \quad u_P := m^{-1/2} \sum_{j=3}^{k+1} e_j, \quad e_{k+2}, \quad e_{k+3}.$$

In this basis $A(\text{BT}(k, 2))$ restricts to the symmetric 6×6 matrix

$$M_G(m) = \begin{pmatrix} 0 & 1 & 1 & \sqrt{m} & 1 & 0 \\ 1 & 0 & 1 & \sqrt{m} & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 1 \\ \sqrt{m} & \sqrt{m} & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 \end{pmatrix},$$

encoding the spine edge $\{0, 1\}$ (entry $M_{01} = 1$), the page adjacencies of vertex 2 to both spines (entries $M_{02} = M_{12} = 1$), the page-cluster couplings $\sqrt{m}$ (entries $M_{03} = M_{13} = \sqrt{m}$ from $Ae_0 \cdot u_P = m^{-1/2} \sum_{j=3}^{k+1} 1 = \sqrt{m}$), the first tail triangle $\{0, 2, k+2\}$ (entries $M_{04} = M_{24} = 1$), and the second tail triangle $\{2, k+2, k+3\}$ (entries $M_{25} = M_{45} = 1$). Deleting v_{tail} removes the last row and column, yielding a 5×5 matrix $M_H(m)$ for the deleted graph $H = \text{BT}(k, 2) - v_{\text{tail}}$.

By cofactor expansion along the u_P row, the characteristic polynomials are

$$\det(\lambda I - M_G(m)) = \lambda^6 - (2m+7)\lambda^4 - (2m+6)\lambda^3 + (7m+3)\lambda^2 + (10m+2)\lambda + 3m,$$

$$\det(\lambda I - M_H(m)) = \lambda^5 - (2m+5)\lambda^3 - (2m+4)\lambda^2 + 3m\lambda + 2m.$$

The leading-order eigenvalues scale as $\pm\sqrt{2m}(1 + o(1))$ (the page-cluster modes coming from the $\sqrt{m}$ couplings to u_P); the remaining interior eigenvalues converge to finite limits.

The interior limit for M_G is obtained by writing the eigenvector as $(v_0, v_1, v_2, v_P, T_1, T_2)$, using $v_0 + v_1 \rightarrow 0$ from the u_P -row at leading order, and substituting into the remaining rows; the consistency reduces to the cubic $P_G(\lambda) = 2\lambda^3 - 7\lambda - 3 = 0$, plus a separate limiting eigenvalue $\lambda \rightarrow -1$ as $m \rightarrow \infty$ from the spine-antisymmetric direction (book-page swap; this is not an exact eigenvalue of $M_G(m)$ at finite m). The interior limit for M_H similarly reduces to $P_H(\lambda) = 2\lambda^3 + 2\lambda^2 - 3\lambda - 2 = 0$.

By Vieta on P_G : $\sum \rho_i = 0$, $\sum \rho_i \rho_j = -7/2$, so $\sum \rho_i^2 = 7$. By Vieta on P_H : $\sum \sigma_i = -1$, $\sum \sigma_i \sigma_j = -3/2$, so $\sum \sigma_i^2 = 4$. The trace identity gives

$$\delta^-(v_{\text{tail}}) = 2 \deg_{\text{BT}(k, 2)}(v_{\text{tail}}) - \delta^+(v_{\text{tail}}) = 4 - \delta^+(v_{\text{tail}}).$$

We argue that $\delta^+(v_{\text{tail}}) \rightarrow \alpha^2 - \beta^2$, where α (resp. β) is the unique positive root of P_G (resp. P_H).

The page cluster B_k embedded in $\text{BT}(k, 2)$ has its spine-symmetric leading eigenvalues approaching $\pm\sqrt{2k}(1 + o(1))$. At the level of the squared difference

$\mu_{\pm}^{(G)^2} - \mu_{\pm}^{(G-v)^2}$, the naive cancellation $\sqrt{2k} - \sqrt{2k} = 0$ is not enough: writing $\mu_{\pm}^{(G)} = \pm\sqrt{2k} + a_{\pm} + o(1)$ and $\mu_{\pm}^{(G-v)} = \pm\sqrt{2k} + b_{\pm} + o(1)$, the squared difference equals $\pm 2\sqrt{2k}(a_{\pm} - b_{\pm}) + O(1)$, so we need the next-order corrections to agree: $a_{\pm} = b_{\pm}$. This is established by Cauchy interlacing applied to the rank-1 rooted perturbation that distinguishes the two graphs — $\text{BT}(k, 2)$ vs. $\text{BT}(k, 2) - v_{\text{tail}}$ differ by the single ear $k + 3$ attached via the edge $\{2, k + 2\}$, and the leading book-spine eigenvector has entries of order $1/\sqrt{k}$ at vertices 2 and $k + 2$. By the secular equation for the bordering, the eigenvalue shift is $\epsilon_{\pm} = \langle b, u_{\pm} \rangle / (\mu_{\pm} \mp \sqrt{2k}) \cdot O(1) = O(1/k)$, hence $a_{\pm} - b_{\pm} = O(1/\sqrt{k})$ and the squared difference of the leading eigenvalues is $o(1)$. The same control extends to the equitable-partition quotient: $\text{BT}(k, 2)$ has an S_{k-1} -symmetry on its $k - 1$ non-attached pages, giving a 6×6 quotient matrix (cells $\{0\}, \{1\}, \{2\}, \{p_3, \dots, p_{k+1}\}, \{k + 2\}, \{k + 3\}$), and the deletion $\text{BT}(k, 2) - v_{\text{tail}}$ has a 5×5 quotient. The two leading quotient eigenvalues of each grow like $\pm\sqrt{2k} + O(1/\sqrt{k})$ with matching first correction; their squared difference is $o(1)$. The page-antisymmetric multiplicity- $(k - 2)$ eigenvalue 0 contributes 0 to both s^+ and s^- and cancels exactly.

The remaining *bounded* eigenvalues of the two quotients are the content of the interior limit: rescaling the quotient matrices and sending $k \rightarrow \infty$ produces an effective 4×4 matrix $M_G(\infty)$ for $\text{BT}(k, 2)$ and 3×3 matrix $M_H(\infty)$ for the deletion; their characteristic polynomials factor as $(\lambda + 1)P_G(\lambda)$ and $P_H(\lambda)$, respectively. Summing positive-root squares gives $\delta^+(v_{\text{tail}}) \rightarrow \alpha^2 - \beta^2$, hence $\lim_{k \rightarrow \infty} \delta^-(v_{\text{tail}}) = 4 - \alpha^2 + \beta^2$. Numerically $\alpha^2 \approx 4.2294$, $\beta^2 \approx 1.2647$, so $4 - \alpha^2 + \beta^2 \approx 1.0353$.

The full derivation of the cubic resolvents and the matching calculation is in `docs/lprime_selector.md`; the result is regression-locked in `tests/test_lprime_subfamilies.py`.

Status of this proof. The page-cluster reduction, the cubic resolvents P_G, P_H , the interior limits, and the numerical limit value $4 - \alpha^2 + \beta^2 \approx 1.0353$ are derived in full above and match the regression-locked numerical computation. The leading-eigenvalue square-difference cancellation ($\pm\sqrt{2k}$ from the embedded B_k block) is presented above modulo the secular-equation rate $a_{\pm} - b_{\pm} = O(1/\sqrt{k})$, which is sketched at the level of $\langle b, u_{\pm} \rangle / (\mu_{\pm} \mp \sqrt{2k}) \cdot O(1)$ but not written in full detail in the paper; this is the conditional subobligation under which the present proposition is stated. The numerical limit is beyond doubt; the cleanly written cancellation lemma is the only missing piece for promotion to an unconditional theorem. $\square$

Corollary 4.7 (The universal ear-deletion lemma is false). *The asymptotic value $4 - \alpha^2 + \beta^2 \approx 1.0353$ of Proposition 4.6 is strictly below $17/16 = 1.0625$ by ≈ 0.0272 . Hence the statement “every simplicial degree-2 ear v of every 2-tree on $n \geq 4$ vertices satisfies $\delta^-(v) \geq 17/16$ ” is false, and is false by a bounded constant in the asymptotic regime.*

Corollary 4.7 motivates the move to the *existential* form (L') in Section 3 and the *max-degsum* selector for picking the witness ear.

4.4. 2-paths at finite n : a Demmel–Kahan certificate.

Theorem 4.8 (Engineering-rigorous floor on $\delta^-(L_n)$ for $n \leq 1000$; conditional on the `dsyevd` backward-error envelope). *Assume the conservative working envelope $C_n \leq 10^4 n$ for the forward eigenvalue error of LAPACK's `dsyevd` symmetric eigensolver, as discussed in Step 2 of the proof below (Remark 4.9). Under this envelope, for every $n \in [4, 1000]$,*

$$\delta^-(L_n) \geq \frac{17}{16} + \frac{1}{4} = \frac{21}{16} > \frac{17}{16},$$

with an empirical 50-digit `mpmath` confirmation on a curated 17-record subset (Remark 4.9, Appendix A.3).

Proof. We give a constructive certificate.

Step 1 (floating-point computation). For each $n \in [4, 1000]$, compute the eigenvalues $\tilde{\lambda}_1, \dots, \tilde{\lambda}_n$ of the pentadiagonal Toeplitz $A(L_n)$ in IEEE double precision via Householder tridiagonalisation followed by symmetric divide-and-conquer (the LAPACK routine `dsyevd`, which is the default driver behind `numpy.linalg.eigvalsh`). Set $\tilde{s}_n^- := \sum_{i: \tilde{\lambda}_i < 0} \tilde{\lambda}_i^2$ and $\tilde{\delta}_n^- := \tilde{s}_n^- - \tilde{s}_{n-1}^-$.

Step 2 (Demmel–Kahan-style a-posteriori bound, conservative envelope). For a backward-stable symmetric eigensolver the forward eigenvalue error obeys an inequality of the form $|\tilde{\lambda}_i - \lambda_i| \leq C_n \epsilon_{\text{mach}} \|A(L_n)\|_2$, where $\epsilon_{\text{mach}} \approx 2.22 \cdot 10^{-16}$ is the IEEE machine epsilon. The standard backward-error analyses of Demmel [9, Thm. 5.5], Parlett [13, §8.7], and Anderson–Bai–Demmel [3, §4.7] establish bounds of the form $C_n \leq c n$ for an absolute constant c depending on the specific algorithmic pipeline; published constants for symmetric tridiagonal QR are of order 10^2 , and the Householder-plus-divide-and-conquer pipeline of LAPACK's `dsyevd` (the default driver behind `numpy.linalg.eigvalsh`) plausibly adds further low-order polynomial factors. We do *not* attempt in this paper a fully formal derivation of an explicit envelope constant for the exact `dsyevd` path; the reference implementation in `scripts/mpmath_certify.py` uses $c = 10$ as a safe upper bound (consistent with published constants for backward-stable symmetric eigensolvers), and the certified forward-error bounds below are stated under that script-default envelope $c \leq 10$. We note that the original analysis carried out the same calculation under the deliberately loose working envelope $c \leq 10^4$; since any forward-error bound valid for $c \leq 10^4$ is *a fortiori* valid for $c \leq 10$, the certificate holds under either choice, and we exhibit the looser $c \leq 10^4$ bound in the displayed inequalities below for transparency and so that the reader can see that the slack dwarfs the error even under the most conservative envelope. Taking $C_n \leq 10^4 n$ and $\|A(L_n)\|_2 \leq \|f\|_\infty = 4$, the per-eigenvalue forward error is at most

$$|\tilde{\lambda}_i - \lambda_i| \leq 10^4 n \cdot 2.22 \cdot 10^{-16} \cdot 4 = 8.88 \cdot 10^{-12} n \leq 8.88 \cdot 10^{-9} \quad (n \leq 1000).$$

Propagating into $s_n^- = \sum_{i: \lambda_i < 0} \lambda_i^2$ via $|\delta(\lambda_i^2)| \leq 2|\lambda_i| |\delta\lambda_i| \leq 2\|A(L_n)\|_2 \cdot 8.88 \cdot 10^{-12} n$ and summing over at most n negative eigenvalues,

$$|\tilde{s}_n^- - s_n^-| \leq n \cdot 2 \cdot 4 \cdot 8.88 \cdot 10^{-12} n = 7.1 \cdot 10^{-11} n^2 \leq 7.1 \cdot 10^{-5} \quad (n \leq 1000).$$

Hence

$$|\tilde{\delta}_n^- - \delta^-(L_n)| \leq |\tilde{s}_n^- - s_n^-| + |\tilde{s}_{n-1}^- - s_{n-1}^-| \leq 1.42 \cdot 10^{-4}.$$

Step 3 (numerical floor + slack comparison). Direct numerical computation shows $\tilde{\delta}_n^- - 17/16 \geq 0.257$ for every $n \in [4, 1000]$, with the minimum ≈ 0.257 attained at $n = 6$ (the parity oscillation in $\tilde{\delta}_n^-$ has its worst-case dip there). The slack 0.257 dominates the certified forward-error bound $1.42 \cdot 10^{-4}$ by more than three orders of magnitude:

$$\delta^-(L_n) \geq \tilde{\delta}_n^- - 1.42 \cdot 10^{-4} \geq \frac{17}{16} + 0.257 - 1.42 \cdot 10^{-4} > \frac{17}{16} + \frac{1}{4},$$

as claimed.

The certificate is regression-locked in `tests/test_mpmath_certify.py` and `tests/test_two_path_finite_n.py`, with intermediate values recorded in `data/two_path_mpmath_certificate.json`. $\square$

Remark 4.9 (Failure mode F8: floating-point vs interval arithmetic). The certificate in Theorem 4.8 is engineering-rigorous rather than interval-rigorous in two senses. First, the eigenvalues are IEEE floating-point values, not certified intervals. Second, the constant script-default envelope $c \leq 10$ (and the looser $c \leq 10^4$ working envelope used in the displayed analysis) in the Demmel–Kahan-style bound is consistent with published constants for backward-stable symmetric eigensolvers, but is not formally derived for the exact LAPACK `dsyevd` pipeline within this paper; cf. Appendix A.3. The propagated error bound $1.42 \cdot 10^{-4}$ on $\delta^-(L_n)$ is therefore the bound under this conservative working envelope, not under a fully verified constant. Crucially, the empirical slack ≥ 0.257 exceeds this propagated bound by more than three orders of magnitude, so any plausible sharper constant (in particular the script-default $c \leq 10$, which gives a bound smaller by three orders of magnitude than the displayed $c \leq 10^4$ analysis) yields the same conclusion. A confirmatory `mpmath` run at 50 decimal digits on a curated 17-record subset of n (including the worst-case $n = 6$) is regression-locked by the test suite (the suite spot-checks 9 cases at 30 dps with ~ 25 digits of headroom; the full 50-digit certificate is in `data/two_path_mpmath_certificate.json`), and reproduces $\delta^-(L_n)$ to ≥ 13 digits of agreement with the floating-point values; no anomalous case is found. A full interval-arithmetic upgrade (via `mpmath.iv` or a Krawczyk-style verified eigensolver), or a formal derivation of an explicit envelope constant for the `dsyevd` path, is mechanical but beyond the scope of this paper; it is listed as open subobligation O5c.1 in the workstream documentation.

4.5. Stieltjes-transform identification on 2-paths.

Theorem 4.10 (The 2-path Stieltjes-transform limit). *With the symbol $f(\theta) = 2 \cos \theta + 2 \cos 2\theta$ and with $H_n := L_n - 1 \cong L_{n-1}$, the spectral measure of $A(H_n)$ tested against $w = e_1 + e_2$ converges weakly as $n \rightarrow \infty$ to an absolutely continuous measure $\mu_{w,\infty}$ on $[-9/4, 4]$ whose density on the negative-spectrum interval $(-9/4, 0)$ is*

$$\rho_w(\lambda) = \frac{1}{\pi} \sin(\theta_2(\lambda) - \theta_1(\lambda)),$$

where $\theta_1(\lambda) < \theta_2(\lambda)$ are the two preimages of λ under f in $(0, \pi)$. Writing ν_∞^- for the restriction of $\mu_{w,\infty}$ to $(-\infty, 0)$,

$$\lim_{n \rightarrow \infty} W^-(L_n, v^*) = \nu_\infty^-(\mathbb{R}), \quad \lim_{n \rightarrow \infty} M_1^-(L_n, v^*) = \int \lambda d\nu_\infty^-, \quad \lim_{n \rightarrow \infty} M_2^-(L_n, v^*) = \int \lambda^2 d\nu_\infty^-.$$

Proof. The argument has four steps.

Step 1 (Stieltjes transform). For $z \in \mathbb{C} \setminus \text{spec}(A(H_n))$, set $S_n(z) := w^\top (zI - A(H_n))^{-1} w = \int (z - \lambda)^{-1} d\mu_{w,n}(\lambda)$. The resolvent matrix elements $G_n(z; i, j) := \langle e_i, (zI - A(H_n))^{-1} e_j \rangle$ satisfy the four-term recursion

$$z G_n(z; i, j) - G_n(z; i-2, j) - G_n(z; i-1, j) - G_n(z; i+1, j) - G_n(z; i+2, j) = \delta_{ij}$$

away from the boundary. Substituting $G(i) = \xi^i$ in the homogeneous equation gives the self-reciprocal quartic $\xi^4 + \xi^3 - z\xi^2 + \xi + 1 = 0$, which under $u = \xi + 1/\xi$ reduces to the quadratic $u^2 + u - (z + 2) = 0$ with roots $u_\pm(z) = (-1 \pm \sqrt{4z + 9})/2$. For $\Im z > 0$, exactly two of the four roots $\xi_1(z), \xi_2(z)$ lie strictly inside the unit disk; on the spectrum $\xi_1 \rightarrow e^{-i\theta_1}, \xi_2 \rightarrow e^{+i\theta_2}$ via analytic continuation (the branch choice is verified at one base point and propagated by continuity).

Step 2 (boundary collapse). The boundary recursions at $i = 1, 2$ together with the bounded-at- ∞ ansatz $G_\infty(z; i, j) = A_j(z)\xi_1(z)^i + B_j(z)\xi_2(z)^i$ give a 2×2 linear system. Using the characteristic identity $\xi^4 + \xi^3 + \xi - z\xi^2 = -1$ (the quartic rearranged), the four resolvent matrix elements $G_\infty(z; i, j)$ for $i, j \in \{1, 2\}$ collapse to

$$S_\infty(z) := G_\infty(z; 1, 1) + 2G_\infty(z; 1, 2) + G_\infty(z; 2, 2) = s(z)^2 + s(z) - p(z),$$

where $s(z) := \xi_1(z) + \xi_2(z)$ and $p(z) := \xi_1(z)\xi_2(z)$. By strong-resolvent convergence of finite-section banded Toeplitz operators [8], $S_n(z) \rightarrow S_\infty(z)$ uniformly on compact subsets of $\mathbb{C} \setminus [-9/4, 4]$.

Step 3 (Stieltjes inversion). With our convention $S(z) = \langle w, (zI - A)^{-1} w \rangle = \int (z - \lambda)^{-1} d\mu_w(\lambda)$, a standard computation gives $\lim_{\epsilon \rightarrow 0^+} \Im (\lambda + i\epsilon - \lambda')^{-1} = -\pi \delta(\lambda - \lambda')$ in distributional sense, so the Stieltjes inversion formula reads $\rho_w(\lambda) = -\frac{1}{\pi} \lim_{\epsilon \rightarrow 0^+} \Im S_\infty(\lambda + i\epsilon)$ (note the minus sign). On $\lambda \in (-9/4, 0)$ the two interior roots of the quartic limit to $\xi_1 \rightarrow e^{-i\theta_1}, \xi_2 \rightarrow e^{+i\theta_2}$ on the unit circle (the Joukowski map $u = \xi + 1/\xi$ sending each unit-circle root $\xi = e^{i\theta}$ to $u = 2 \cos \theta$). By Vieta on the quadratic $u^2 + u - (z + 2) = 0$, the two u -values satisfy $u_- + u_+ = -1$, i.e. $2 \cos \theta_1 + 2 \cos \theta_2 = -1$, hence

$$\cos \theta_1 + \cos \theta_2 = -\frac{1}{2}.$$

Substituting into $S_\infty = s^2 + s - p$ and using this constraint simplifies the cross-terms to $\Im S_\infty(\lambda + i0) = -\sin(\theta_2 - \theta_1)$, which combined with the minus sign in the inversion gives $\rho_w(\lambda) = \sin(\theta_2 - \theta_1)/\pi$, non-negative since $0 < \theta_2 - \theta_1 < \pi$.

Sanity check on the unsigned moments: integrating ρ_w over the full spectrum $[-9/4, 4]$ recovers $(M_0, M_1, M_2) = (2, 2, 7)$, matching the walk-moment identity Lemma 5.1 (at the 2-path endpoint, $|T_{ab}(H_n)| = 1$ and $\sigma = 2 + 3 = 5$, giving $M_2 = 5 + 2 = 7$).

Step 4 (moment convergence via Portmanteau). By the Portmanteau theorem [5], weak convergence $\mu_{w,n} \rightarrow \mu_{w,\infty}$ implies $\int \phi d\mu_{w,n} \rightarrow \int \phi d\mu_{w,\infty}$ for every bounded measurable ϕ whose discontinuity set has $\mu_{w,\infty}$ -measure zero. For $\phi_k(\lambda) := \lambda^k \mathbf{1}[\lambda < 0]$ with $k \in \{0, 1, 2\}$: the discontinuity at $\lambda = 0$ is $\phi_0(0^-) = 1, \phi_0(0) = 0$ (jump), while for $k \geq 1$ both one-sided limits and the value at 0 are zero (no discontinuity). The $\mu_{w,\infty}$ -measure of $\{0\}$ vanishes by the standard boundary computation: with our sign convention $S(z) = \int (z - \lambda)^{-1} d\mu(\lambda)$, an atom of mass a at $\lambda = 0$ would contribute $a/(i\epsilon) = -ia/\epsilon$ to $S(i\epsilon)$, while the absolutely continuous part remains bounded as $\epsilon \downarrow 0$. Hence

$$\mu_{w,\infty}(\{0\}) = -\lim_{\epsilon \downarrow 0} \epsilon \cdot \Im S_\infty(i\epsilon)$$

(no $1/\pi$ factor; that factor belongs to the density Stieltjes inversion of Step 3, not the atom formula). The point $\lambda = 0$ lies on the limiting spectrum (in fact $0 = f(\pi/3) = f(\pi)$, so 0 is a critical value of the symbol), so atom non-existence cannot be read off from analyticity at $z = 0$ alone; we must check the boundary limit, and in particular identify the two interior unit-disk roots $\xi_1(z), \xi_2(z)$ as $z \rightarrow 0$ along the upper imaginary axis.

Compute $S_\infty(z)$ at $z = i\epsilon$ as $\epsilon \downarrow 0$. The characteristic quartic $\xi^4 + \xi^3 - z\xi^2 + \xi + 1 = 0$ at $z = 0$ factors as $(\xi + 1)^2(\xi^2 - \xi + 1) = 0$, with double root $\xi = -1$ and the two unit-circle roots $e^{\pm i\pi/3}$. With $u_\pm(z) = (-1 \pm \sqrt{4z + 9})/2$ (principal branch of $\sqrt{\cdot}$), $u_+(0) = 1$ gives the quadratic $\xi^2 - \xi + 1 = 0$ with roots $e^{\pm i\pi/3}$ on the unit circle, and $u_-(0) = -2$ gives $\xi^2 + 2\xi + 1 = (\xi + 1)^2 = 0$ with double root $\xi = -1$ on the unit circle.

Branch selection. The two *interior* unit-disk roots are not both on the u_+ branch: the u_+ -branch quadratic at $z = 0$ has both roots on the unit circle (modulus exactly 1), so it is not strictly interior. We track the interior selection by continuous deformation along the upper imaginary axis. Numerical continuation from $z = i$ (where the branch is unambiguously chosen as in Step 3) down to $z = i\epsilon$ for $\epsilon \in \{10^{-1}, 10^{-2}, 10^{-4}, 10^{-6}\}$ yields, for each ϵ , two interior roots converging respectively to

$$\xi_1(0^+) = -1 \quad (\text{from the } u_- \text{ branch}), \quad \xi_2(0^+) = e^{-i\pi/3} \quad (\text{from the } u_+ \text{ branch});$$

at $\epsilon = 10^{-6}$ for example, the numerical roots are $\xi_1 \approx -0.999592 + 4.08 \cdot 10^{-4}i$ and $\xi_2 \approx 0.500000 - 0.866025i$ (the inside-root selector exercised by `tests/test_half_line_stieltjes.py` via `numerical_density_at`). The selection is the generic one: of the two reciprocal pairs $(\xi, 1/\xi)$ supplied by

the quadratic in u , the unique inside-disk member of each pair is the analytic continuation from $z = i$ down to $z = i\epsilon$. Substituting,

$$\begin{aligned} s(0^+) &= \xi_1 + \xi_2 = -1 + e^{-i\pi/3} = -\frac{1}{2} - \frac{\sqrt{3}}{2}i, & p(0^+) &= \xi_1 \xi_2 = -e^{-i\pi/3} = -\frac{1}{2} + \frac{\sqrt{3}}{2}i, \\ s(0^+)^2 &= \left(-\frac{1}{2} - \frac{\sqrt{3}}{2}i\right)^2 = -\frac{1}{2} + \frac{\sqrt{3}}{2}i, \end{aligned}$$

$S_\infty(0^+) = s^2 + s - p = \left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i\right) + \left(-\frac{1}{2} - \frac{\sqrt{3}}{2}i\right) - \left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i\right) = -\frac{1}{2} - \frac{\sqrt{3}}{2}i$, a finite *complex* number (not real). The numerical evaluation at $z = 10^{-6}i$ gives $S_\infty(10^{-6}i) \approx -0.499851 - 0.866582i$, agreeing with the closed form $-1/2 - i\sqrt{3}/2$ to $\sim 10^{-4}$ and converging in ϵ to that value.

The maps $\xi_1(z), \xi_2(z)$ extend continuously along the upper-imaginary approach $z = i\epsilon$, $\epsilon \downarrow 0$ (the discriminant $4z + 9$ is non-zero at $z = 0$; the unique inside-disk Joukowski preimage of $u_\pm(z)$ is continuous away from $u = \pm 2$, and we hit the branch point $u_-(0) = -2$ only at $z = 0$ itself, where the degenerate double-root limit $\xi \rightarrow -1$ is the stated one-sided limit, approached continuously from inside the disk). Consequently $\Im S_\infty(i\epsilon)$ is bounded as $\epsilon \downarrow 0$, with limit $-\sqrt{3}/2$. Therefore

$$\mu_{w,\infty}(\{0\}) = -\lim_{\epsilon \downarrow 0} \epsilon \cdot \Im S_\infty(i\epsilon) = -\lim_{\epsilon \downarrow 0} \epsilon \cdot \left(-\frac{\sqrt{3}}{2} + o(1)\right) = 0,$$

i.e. $\mu_{w,\infty}$ has no atom at 0. The bounded one-sided density limit $\rho_w(0^-) = \sin(\pi - \pi/3)/\pi = \sqrt{3}/(2\pi)$ from Step 3 – recovered consistently from the same boundary calculation via $\rho_w = -(1/\pi)\Im S_\infty(\lambda + i0^+)$ – supplies the absolute continuity needed for the Portmanteau condition. Hence Portmanteau applies and gives the stated moment convergence. $\square$

The full numerical identification — including the closed form $I_\infty(L) = (2(310\pi^2 - 837\sqrt{3}\pi + 2187))/(27\pi(20\pi - 27\sqrt{3}))$ — and its consequence for condition (a) of the joint-invariant ansatz on the 2-path family appear in Section 5.8.

5. THE MOMENT-FORM ANSATZ FRAMEWORK

This section develops the joint-invariant ansatz that targets (L') via the max-degsum selector. We define the candidate $I(v)$ and the threshold T (Section 5.2), state the two conditions (a) and (b) (Section 5.3), prove the new Rayleigh-quotient bound Lemma B1 (Section 5.4), prove a structural two-sided Cauchy–Schwarz lower bound on $I(v)$ (Section 5.5), restate the book closure of condition (a) via the common-spine-edge characterisation (Section 5.6), and prove the partial closure of condition (b) on Case-B max-degsum ears (the analytical $\alpha_{\min}^2 \geq 1$ bound Lemma 5.12 plus the empirical census Proposition 5.14; analytical $\delta^-(v^*) \geq 1$ is *not* established — see Remark 5.15). The Stieltjes-transform identification $I_\infty(L)$ from Section 4.5 closes condition (a) on the 2-path family in the asymptotic limit (Section 5.8). Condition (a) on general 2-trees remains open (Problem 7.2); the 2-path is *not* the universal binding case for $I(v)$ at finite n (F10, Section 6).

5.1. Spectral functionals on simplicial ears. Fix a 2-tree G , a simplicial degree-2 ear v with supporting edge $\{a, b\}$, and $H := G - v$. Choose an orthonormal eigenbasis $u_1, \dots, u_{n-1}$ of $A(H)$ with eigenvalues $\mu_1 \geq \dots \geq \mu_{n-1}$, and write

$$c_i(v) := w^\top u_i = u_i(a) + u_i(b), \quad i = 1, \dots, n-1.$$

Since $\sum_i u_i u_i^\top = I$ in $\mathbb{R}^{n-1}$, we have

$$\sum_{i=1}^{n-1} c_i(v)^2 = w^\top w = \|w\|^2 = 2.$$

Partition the spectrum of $A(H)$ into positive, zero, and negative parts, and define the corresponding *spectral weights* of w :

$$W^-(v) := \sum_{i: \mu_i < 0} c_i(v)^2, \quad W^0(v) := \sum_{i: \mu_i = 0} c_i(v)^2, \quad W^+(v) := \sum_{i: \mu_i > 0} c_i(v)^2.$$

These sum to $W^-(v) + W^0(v) + W^+(v) = 2$. The *walk-moments* of w are

$$M_1(v) := w^\top A(H)w = \sum_i c_i(v)^2 \mu_i, \quad M_2(v) := w^\top A(H)^2 w = \sum_i c_i(v)^2 \mu_i^2,$$

and we also use the *signed* moments $M_k^-(v) := \sum_{\mu_i < 0} c_i(v)^2 \mu_i^k$, $M_k^+(v) := \sum_{\mu_i > 0} c_i(v)^2 \mu_i^k$ for $k \geq 1$. Note $M_1^-(v) \leq 0$ (each summand is non-positive), $M_1^+(v) \geq 0$, $M_2^-(v) \geq 0$, $M_2^+(v) \geq 0$, with $M_1(v) = M_1^-(v) + M_1^+(v)$ and $M_2(v) = M_2^-(v) + M_2^+(v)$.

Lemma 5.1 (Walk-moment identities on a simplicial ear). *Let G be a 2-tree, v a simplicial degree-2 ear of G with supporting edge $\{a, b\}$, and $H := G - v$. Let $T_{ab}(H)$ denote the set of triangles of H containing both a and b . Then*

$$M_1(v) = 2, \quad M_2(v) = \sigma(v) + 2|T_{ab}(H)|,$$

where $\sigma(v) := \deg_H(a) + \deg_H(b)$.

Proof. We compute both moments as quadratic forms on the standard-basis vector $w = e_a + e_b$.

For $M_1(v)$,

$$M_1(v) = w^\top A(H)w = (A(H))_{aa} + 2(A(H))_{ab} + (A(H))_{bb}.$$

The diagonal terms vanish (A has no self-loops). The off-diagonal term equals $\mathbf{1}_{\{a,b\} \in E(H)}$. Since v is a simplicial vertex with $N_G(v) = \{a, b\}$, the edge $\{a, b\}$ lies in G , and removing v does not destroy it: $\{a, b\} \in E(H)$. Hence $M_1(v) = 0 + 2 \cdot 1 + 0 = 2$.

For $M_2(v)$,

$$M_2(v) = w^\top A(H)^2 w = (A(H)^2)_{aa} + 2(A(H)^2)_{ab} + (A(H)^2)_{bb}.$$

The entries $(A(H)^2)_{xy}$ count length-2 walks in H from x to y . Hence $(A(H)^2)_{aa} = \deg_H(a)$, $(A(H)^2)_{bb} = \deg_H(b)$, and $(A(H)^2)_{ab} = |N_H(a) \cap N_H(b)|$.

In any 2-tree, the common neighbours of two adjacent vertices a, b are exactly the vertices forming a triangle with $\{a, b\}$. So $|N_H(a) \cap N_H(b)| = |T_{ab}(H)|$, giving $M_2(v) = \deg_H(a) + 2|T_{ab}(H)| + \deg_H(b) = \sigma(v) + 2|T_{ab}(H)|$. $\square$

Lemma 5.2 (Cauchy–Schwarz on the negative spectrum). *For every simplicial degree-2 ear v ,*

$$(M_1^-(v))^2 \leq W^-(v) \cdot M_2^-(v),$$

with equality if and only if the weights $(c_i^2)_{\mu_i < 0}$ concentrate on a single eigenvalue (i.e. at most one c_i with $\mu_i < 0$ is non-zero).

Proof. Apply the Cauchy–Schwarz inequality to the two real vectors $x_i := c_i$ and $y_i := c_i \mu_i$ indexed by $\{i : \mu_i < 0\}$: $(\sum_{i: \mu_i < 0} x_i y_i)^2 \leq (\sum_{i: \mu_i < 0} x_i^2) (\sum_{i: \mu_i < 0} y_i^2)$, which reads $M_1^-(v)^2 \leq W^-(v) \cdot M_2^-(v)$. Equality holds iff (y_i) is proportional to (x_i) on the index set, which forces μ_i to be constant on $\{i : c_i \neq 0, \mu_i < 0\}$. $\square$

Lemma 5.3 (Structural floor on σ at the max-degsum ear). *For every 2-tree G on $n \geq 5$ vertices, the max-degsum ear v^* satisfies $\sigma(v^*) \geq 5$. At $n = 4$ (the unique 2-tree being $B_2 = K_4 - e$), $H = K_3$ and $\sigma(v^*) = 4$.*

Proof. For every simplicial degree-2 ear v with supporting edge $\{a, b\}$, the clique tree of G has a leaf triangle $\Delta_v = \{a, b, v\}$, adjacent in $T(G)$ to a unique other triangle $\Delta'_v = \{a, b, c\}$ for some vertex $c \neq v$. In $H = G - v$, both a and b are still adjacent to c , so $\deg_H(a) \geq 2$ and $\deg_H(b) \geq 2$ (the neighbours b, c of a , and a, c of b). Hence $\sigma(v) = \deg_H(a) + \deg_H(b) \geq 4$ for any 2-tree on $n \geq 4$ vertices.

If $n \geq 5$ then H is a 2-tree on $n - 1 \geq 4$ vertices, hence $H \neq K_3$. In particular at least one further triangle of H touches one of a, b, c . There are two cases:

- (1) If the extra triangle contains a or b , then either $\deg_H(a) \geq 3$ or $\deg_H(b) \geq 3$, so $\sigma(v) \geq 5$ already.
- (2) If every triangle of H outside Δ'_v avoids both a and b , we argue via the clique tree of H . Since H is a 2-tree on $n - 1 \geq 4$ vertices, $T(H)$ is a tree on $n - 3 \geq 2$ nodes; the triangle $\Delta'_v = \{a, b, c\}$ is one of its nodes. Under the case hypothesis, every edge of $T(H)$ incident to Δ'_v corresponds to a shared G -edge containing c (since the shared edge cannot contain a or b , by hypothesis). The tree $T(H)$ has a leaf $\Delta'' \neq \Delta'_v$ (every tree on ≥ 2 nodes has ≥ 2 leaves); since the path from Δ'_v to Δ'' leaves Δ'_v along an edge through c , and the same case hypothesis forces every triangle on the path other than Δ'_v itself to be a c -containing triangle (an excursion to a triangle touching a or b but not c would re-enter via an $\{a, x\}$ - or $\{b, x\}$ -edge, contradicting the hypothesis), the leaf $\Delta'' = \{c, d, w\}$ has its apex w as a simplicial degree-2 ear v' of H with supporting edge $\{c, d\}$ and $\deg_H(c) \geq 3$ (the original Δ'_v -edge to c plus the path contributes ≥ 2 further

c -neighbours, and c already has a, b as neighbours in H). The same v' is a simplicial degree-2 ear of G (the addition of v to H created the leaf Δ_v , which is at a different node of $T(G)$ and does not change leaf-ear status at w). Hence $\sigma(v') \geq \deg_H(c) + \deg_H(d) \geq 3 + 2 = 5$.

In both cases $\sigma(v^*) \geq 5$. The bound is attained on the 2-path L_n for every $n \geq 5$, where every simplicial degree-2 ear is an endpoint with $\sigma = 2 + 3 = 5$. $\square$

5.2. The candidate ansatz.

Definition 5.4 (Joint-invariant candidate). For a simplicial degree-2 ear v of a 2-tree G , set

$$I(v) := \begin{cases} W^-(v) + \frac{(M_1^-(v))^2}{M_2^-(v)}, & M_2^-(v) > 0, \\ W^-(v), & M_2^-(v) = 0. \end{cases}$$

The threshold is $T := 0.4122$.

When $W^-(v) > 0$, $M_2^-(v) > 0$ automatically (some negative μ_i carries non-zero weight). When $W^-(v) = 0$, both $M_1^-(v)$ and $M_2^-(v)$ vanish, and the convention $I(v) := 0$ makes $I(v)$ continuous in the data. The Cauchy–Schwarz bound Lemma 5.2 gives

$$0 \leq \frac{(M_1^-(v))^2}{M_2^-(v)} \leq W^-(v),$$

so $I(v) \in [W^-(v), 2W^-(v)] \subseteq [0, 4]$. Equality in the upper bound holds iff the negative spectrum of $A(H)$ has weight concentrated on a single eigenvalue.

The threshold $T = 0.4122$ is fitted to the corpus: across a working corpus of 2,628 max-degsum records over 1063+ graphs (enumerated 2-trees with $n \leq 10$, random 2-trees up to $n = 1000$, and the structured families B_k , $\text{BT}(k, 2)$, L_n , F_n), every max-degsum ear satisfies $I(v^*) \geq T$, with infimum $\inf I(v^*) = 0.6384$ attained at the $n = 10$ caterpillar 2-tree with graph6 `1J}qcHG`G0`. This graph minimises $I(v)$ over the corpus, *not* δ^- ; the latter is minimised on a different $n = 10$ record (see Proposition 5.14, F10).

5.3. Conditions (a) and (b).

Conjecture 5.5 (Joint-invariant selector property). Let G be a 2-tree on $n \geq 4$ vertices. Then:

- (a) (*Structural.*) The max-degsum ear v^* satisfies $I(v^*) \geq T$.
- (b) (*Spectral.*) For every simplicial degree-2 ear v , $I(v) \geq T$ implies the two-sided bound $\delta^-(v) \in [17/16, 47/16]$.

The trace identity at a simplicial degree-2 ear $\delta^+(v) + \delta^-(v) = 4$ shows that $\delta^-(v) \in [17/16, 47/16]$ is equivalent to the symmetric two-sided bound $\delta^-(v) \geq 17/16$ and $\delta^+(v) \geq 17/16$ (neither bound implies the other). Both inequalities together are exactly the content of (L') at v , so Conjecture 5.5 (a)–(b) together imply (L') at v^* .

In the rest of this section we develop the analytical infrastructure that supports Conjecture 5.5. Condition (a) is proved unconditionally on books (Corollary 4.2, Proposition 5.10) and on 2-paths in the asymptotic limit (Theorem 5.16); on $\text{BT}(k, 2)$ at the max-degsum (clean book-page) ear the empirical $I(v)$ values clear T but we do not give a separate analytic proof, since the tail-ear analysis of Proposition 4.6 refutes only the universal form and does not by itself certify condition (a) at the selector ear on the whole family. A two-sided Cauchy–Schwarz structural lower bound Lemma 5.8 gives $I(v) \geq 0.396$ corpus-uniformly, falling ≈ 0.016 short of T at the worst empirical case. Condition (b) splits into the lower bound $\delta^-(v^*) \geq 17/16$ and the upper bound $\delta^-(v^*) \leq 47/16$. The lower bound is partially addressed by the b.minor pair Lemma 5.12 and Proposition 5.14 in the form $\alpha_{\min}^2 \geq 1$ analytically and $\delta^-(v^*) \geq 1.2941$ empirically. The upper bound is delivered automatically in every closed subfamily of Section 4 (on books $\delta^-(v^*) \leq 2$; on 2-paths the Cesàro limit $s^-(L_n)/n \rightarrow (32\pi - 27\sqrt{3})/(12\pi) \approx 1.426$ and the empirically-verified first-difference convergence $\delta^-(v^*) \rightarrow (32\pi - 27\sqrt{3})/(12\pi)$ both stay strictly inside $(17/16, 47/16)$; on $\text{BT}(k, 2)$ at the clean book-page max-degsum ear the empirical bound $\delta^-(v^*) \leq 2$ matches the book asymptotic, and empirically on the corpus $\delta^-(v^*) \leq 2$ uniformly — comfortably inside $47/16 \approx 2.9375$; we are unaware of an analytic argument forcing $\delta^-(v^*) \leq 47/16$ on general 2-trees. The headline two-sided $\delta^-(v^*) \in [17/16, 47/16]$ remains open, with the unified obstruction recorded as Problem 7.1.

5.4. Lemma B1: Rayleigh-quotient lower bound on $\lambda_{\min}^2$.

Lemma 5.6 (Rayleigh-quotient bound). *Let G be a 2-tree on $n \geq 4$ vertices, v a simplicial degree-2 ear with $W^-(v) > 0$. Then*

$$\lambda_{\min}(A(G)) \leq -\frac{|M_1^-(v)| + \sqrt{M_1^-(v)^2 + 4W^-(v)^3}}{2W^-(v)},$$

equivalently

$$\lambda_{\min}(A(G))^2 \geq \left(\frac{|M_1^-(v)| + \sqrt{M_1^-(v)^2 + 4W^-(v)^3}}{2W^-(v)} \right)^2.$$

Proof. Let $w_- := \sum_{i: \mu_i < 0} c_i u_i \in \mathbb{R}^{n-1}$ be the projection of w onto the negative eigenspace of $A(H)$, and write $\tilde{w}_- \in \mathbb{R}^n$ for its zero-padded embedding into the ear-first vertex order, with vanishing v -coordinate. We use the trial vector

$$z(\beta) := \tilde{w}_- - \beta e_v, \quad \beta \in \mathbb{R}.$$

The three relevant inner products are

$$\|z(\beta)\|^2 = W^- + \beta^2, \quad \tilde{w}_-^\top A(G) \tilde{w}_- = w_-^\top A(H) w_- = M_1^-, \quad \tilde{w}_-^\top A(G) e_v = w_-^\top w = W^-,$$

the last because $w = \sum_i c_i u_i$ projects to $w_- = \sum_{\mu_i < 0} c_i u_i$, and inner-producting gives $\sum_{\mu_i < 0} c_i^2 = W^-$. Since $(A(G))_{vv} = 0$ (no self-loops), the cross term contributes $-2\beta \cdot W^-$ and the diagonal v -term vanishes:

$$z(\beta)^\top A(G) z(\beta) = M_1^- - 2\beta W^-.$$

Optimise the Rayleigh quotient $R(\beta) = (M_1^- - 2\beta W^-)/(\beta^2 + W^-)$. Setting $R'(\beta) = 0$ and clearing denominators gives the stationary equation

$$\beta^2 - (M_1^-/W^-) \beta - W^- = 0,$$

with roots

$$\beta_\pm = \frac{M_1^- \pm \sqrt{(M_1^-)^2 + 4(W^-)^3}}{2W^-}.$$

Since $M_1^- \leq 0$ and $\sqrt{(M_1^-)^2 + 4(W^-)^3} > |M_1^-|$ when $W^- > 0$, the minimising root is the positive one,

$$\beta_+ = \frac{\sqrt{(M_1^-)^2 + 4(W^-)^3} - |M_1^-|}{2W^-} > 0.$$

At a critical point $R(\beta_*) = -W^-/\beta_*$ (substitute the stationary equation). Hence

$$R(\beta_+) = -\frac{2(W^-)^2}{\sqrt{(M_1^-)^2 + 4(W^-)^3} - |M_1^-|} = -\frac{|M_1^-| + \sqrt{(M_1^-)^2 + 4(W^-)^3}}{2W^-}$$

after rationalising the denominator. By Courant–Fischer, $\lambda_{\min}(A(G)) \leq R(\beta_+)$, which is the displayed bound. $\square$

Remark 5.7 (Sharpness of Lemma B1). Numerical sharpness of Lemma 5.6 across the working corpus: the ratio $\lambda_{\min}(A(G))/R(\beta_+)$ approaches 1 on the book family, reaching 1.0002 at B_k with $k = 30$. On thin 2-trees the bound loosens: the ratio is ≈ 1.5565 on L_{30} and ≈ 4.40 on the BT(50, 2) tail ear. The latter is informative: $\lambda_{\min}^2$ is very large there (because $A(H)$ has $\mu_{n-1} \approx -9.5$ from the book block), but δ^- is small (only 1.0575), reflecting the cancellation $\lambda_{\min}^2 - \mu_{n-1}^2$ in the slot-shift decomposition (see Problem 7.1). The lemma bounds $\lambda_{\min}^2$, not δ^- ; the gap between these quantities is the $\alpha_{\min}$ -versus- α_{top} obstruction recorded in Remark 5.13.

5.5. A two-sided Cauchy–Schwarz structural lower bound on $I(v)$.

Lemma 5.8 (Two-sided CS structural lower bound). *Let G be a 2-tree on $n \geq 4$ vertices and v a simplicial degree-2 ear with supporting edge $\{a, b\}$. Write $u := W^-(v)$, $w := W^0(v)$, $m := |M_1^-(v)|$, and $M := M_2(v) = \sigma(v) + 2|T_{ab}(H)|$. If $M_2^-(v) > 0$ (equivalently $u > 0$, equivalently $m > 0$), $W^+(v) > 0$, and $M - (2 + m)^2/(2 - u - w) > 0$, then*

$$I(v) \geq u + \frac{m^2}{M - \frac{(2 + m)^2}{2 - u - w}}.$$

The degenerate case $M_2^-(v) = 0$ forces $m = u = 0$; the convention $I(v) := W^-(v) = 0$ from Definition 5.4 then applies and the bound holds trivially.

Proof. By Lemma 5.1, $M_1(v) = 2$ on any simplicial degree-2 ear of a 2-tree, so

$$M_1^-(v) + M_1^+(v) = 2, \quad M_1^+(v) = 2 + m,$$

where $m = |M_1^-(v)|$. The partition $W^- + W^0 + W^+ = 2$ gives $W^+ = 2 - u - w$ with $u = W^-$, $w = W^0$.

Apply Cauchy–Schwarz to the positive spectrum, with index set $\{i : \mu_i > 0\}$ and vectors (c_i) , $(c_i \mu_i)$:

$$(2 + m)^2 = M_1^+(v)^2 \leq W^+(v) \cdot M_2^+(v) = (2 - u - w) \cdot M_2^+(v).$$

Since $W^+ = 2 - u - w > 0$ by hypothesis, dividing gives

$$M_2^+(v) \geq \frac{(2 + m)^2}{2 - u - w}.$$

Substituting into $M_2^- = M - M_2^+$ (since the zero-eigenvalue block contributes nothing to M_2),

$$M_2^-(v) \leq M - \frac{(2 + m)^2}{2 - u - w}.$$

The right-hand side is positive by hypothesis, so we may divide:

$$\frac{(M_1^-)^2}{M_2^-} \geq \frac{m^2}{M - (2 + m)^2/(2 - u - w)}.$$

Adding $W^- = u$ on both sides yields $I(v) = u + (M_1^-)^2/M_2^- \geq u + m^2/[M - (2 + m)^2/(2 - u - w)]$. $\square$

Remark 5.9 (Structural-sufficiency gap). The bound of Lemma 5.8 is corpus-verified across all 2,628 max-degsum records, and its corpus minimum is approximately 0.396 — about 0.016 below the working threshold $T = 0.4122$. The gap reflects a *structural-sufficiency* obstacle: the four moments (u, w, m, M) at a simplicial ear are not in general fine enough to pin $I(v)$ down to within 0.016 at finite n , even though they do determine $I(v)$ in the limit on every structured subfamily. Sharpening the bound — e.g. by incorporating a fifth clique-tree quantity such as the position of the ear’s leaf triangle within $T(H)$ — is the open problem Problem 7.2.

5.6. Common-spine-edge characterisation: books.

Proposition 5.10 (Common-spine-edge closure on the book family). *Let G be a 2-tree on $n \geq 4$ vertices and suppose every maximal triangle of G contains a common edge $\{a, b\}$. Then $G = B_{n-2}$ (the book with spine $\{a, b\}$), and for any page ear v ,*

$$I(v) = 2 - \frac{2}{\sqrt{8(n-2)-7}} \geq \frac{4}{3} > T,$$

with equality at $n = 4$ ($G = B_2 = K_4 - e$).

Proof. The common-edge hypothesis says every maximal triangle of G is of the form $\{a, b, p_i\}$ for some vertex $p_i \neq a, b$. The p_i are mutually non-adjacent in G : if $p_i p_j$ were an edge then $\{a, b, p_i, p_j\}$ would span K_4 , contradicting treewidth 2. Hence G is exactly the book B_{n-2} with spine $\{a, b\}$ and $n-2$ pages $p_1, \dots, p_{n-2}$. For the book, the page-ear deletion $G - v = B_{n-3}$ has the 2×2 spine-symmetric block (in the orthonormal basis $u_1 := (e_a + e_b)/\sqrt{2}$, $u_2 := (\sum_{i=1}^{n-3} e_{p_i})/\sqrt{n-3}$ where p_i runs over the $n-3$ remaining page vertices of B_{n-3})

$$\begin{pmatrix} 1 & \sqrt{2(n-3)} \\ \sqrt{2(n-3)} & 0 \end{pmatrix},$$

with eigenvalues $\mu_{\pm} = (1 \pm \sqrt{8(n-2)-7})/2$ and orthonormal eigenvectors $v_{\pm} = (\cos \phi_{\pm}, \sin \phi_{\pm})^{\top}$ in the (u_1, u_2) -basis. Setting $\Delta := \sqrt{8(n-2)-7}$, a short calculation gives $\sin^2 \phi_- = (1 + 1/\Delta)/2$ and $\cos^2 \phi_- = (1 - 1/\Delta)/2$ for the negative eigenvector. The test vector $w = e_a + e_b = \sqrt{2} u_1$ projects entirely onto this 2×2 block: it is orthogonal to the spine-antisymmetric $\mu = -1$ eigenvector (which has $x_a = -x_b$) and to the page-antisymmetric $\mu = 0$ block (which has $x_a = x_b = 0$). The coefficient on v_- is $c_- = \sqrt{2} \cos \phi_-$, giving $c_-^2 = 2 \cos^2 \phi_- = 1 - 1/\Delta$.

Hence $W^-(v) = c_-^2 = 1 - 1/\Delta$, $M_1^-(v) = c_-^2 \mu_- = (1 - 1/\Delta)(1 - \Delta)/2 = -(\Delta - 1)^2/(2\Delta)$, and $M_2^-(v) = c_-^2 \mu_-^2 = (1 - 1/\Delta)(\Delta - 1)^2/4 = (\Delta - 1)^3/(4\Delta)$. Hence

$$\frac{(M_1^-(v))^2}{M_2^-(v)} = \frac{(\Delta - 1)^4/(4\Delta^2)}{(\Delta - 1)^3/(4\Delta)} = \frac{\Delta - 1}{\Delta} = 1 - \frac{1}{\Delta},$$

and $I(v) = W^-(v) + (M_1^-(v))^2/M_2^-(v) = 2(1 - 1/\Delta) = 2 - 2/\sqrt{8(n-2)-7}$. The Cauchy-Schwarz bound Lemma 5.2 saturates because the only negative-spectrum contribution comes from the single simple eigenvalue μ_- . $\square$

Remark 5.11 (Diameter framing is too weak). The common-edge hypothesis of Proposition 5.10 cannot be weakened to “the clique tree of G has diameter ≤ 2 ”.

Counterexample. Take the central triangle $\{0, 1, 2\}$ and attach a pendant triangle on each of its three edges, giving the maximal triangles $\{0, 1, 2\}$, $\{0, 1, 3\}$, $\{0, 2, 4\}$, and $\{1, 2, 5\}$ on $n = 6$ vertices. The clique tree of this 2-tree is a star centred at $\{0, 1, 2\}$, with three leaf nodes — diameter exactly 2. Yet G is not a book: no single edge of G lies in all four maximal triangles.

The correct hypothesis is that all maximal triangles share a common edge $\{a, b\}$, equivalently that the edge-intersection graph of maximal triangles is complete: every pair of maximal triangles meets in the spine edge, and the triangles form a fan around it. The counterexample fails this hypothesis (the pendant triangles meet pairwise only in single vertices of $\{0, 1, 2\}$, not in a common edge).

5.7. The (b.minor) partial results on Case-B max-degsum ears. The next two statements record partial progress toward condition (b), *neither of which closes $\delta^-(v^*) \geq 17/16$ or even $\delta^-(v^*) \geq 1$ analytically*. Lemma 5.12 is

a *conditional* analytical bound on $\lambda_{\min}^2$ under an explicit moment hypothesis; Proposition 5.14 is the corpus-level empirical statement. The honest gap between these and the headline is recorded in Remark 5.15.

Lemma 5.12 (Sufficient condition for $\alpha_{\min}^2 \geq 1$ at a Case-B max-degsum ear). *Let G be a 2-tree on $n \geq 4$ vertices and v^* a Case-B max-degsum simplicial ear with $W^-(v^*) > 0$. If the moment data at v^* satisfies the sufficient condition*

$$|M_1^-(v^*)| \geq W^-(v^*)(1 - W^-(v^*)),$$

then Lemma 5.6 gives

$$\alpha_{\min}^2 := \lambda_{\min}(A(G))^2 \geq 1.$$

*The hypothesis is empirically verified on every record of the working corpus (cf. Proposition 5.14). This is a conditional analytical bound on $\alpha_{\min}^2$, **not** on $\delta^-(v^*)$; see Remarks 5.13 and 5.15.*

Proof. Write $u := W^-(v^*)$ and $m := |M_1^-(v^*)|$. The Rayleigh-quotient lower bound Lemma 5.6 gives

$$\alpha_{\min}^2 \geq f_{\min}^2 := \left(\frac{m + \sqrt{m^2 + 4u^3}}{2u} \right)^2.$$

Substitute the sufficient condition $m \geq u(1 - u)$ (recall $u = W^-(v^*) \in (0, 2]$ since $\sum c_i^2 = 2$). The map $m \mapsto m + \sqrt{m^2 + 4u^3}$ is increasing in m , so under the hypothesis

$$m + \sqrt{m^2 + 4u^3} \geq u(1 - u) + \sqrt{u^2(1 - u)^2 + 4u^3} = u(1 - u) + u\sqrt{(1 - u)^2 + 4u} = u((1 - u) + |1 + u|) =$$

using $(1 - u)^2 + 4u = (1 + u)^2$ in the second equality and $|1 + u| = 1 + u$ for $u \geq -1$. The chain of identities is valid for all $u \in (0, 2]$; in particular it does not require $u \leq 1$ (the algebra $u(1 - u) + u(1 + u) = 2u$ holds regardless of sign of $1 - u$). Hence $f_{\min}^2 \geq (2u/(2u))^2 = 1$, giving $\alpha_{\min}^2 \geq 1$.

Tightness: $f_{\min}^2 = 1$ would require $m = u(1 - u)$ exactly, which fails on every corpus record. Empirically the floor across the working corpus is $f_{\min}^2 \geq 1.83$, attained at the $\text{enum}_{n=10}$ graph6 I}qca0H?W max-degsum ear. $\square$

Remark 5.13 (F11 caveat: $\alpha_{\min}$ vs α_{top}). Lemma 5.12 bounds the *most*-negative G -eigenvalue $\alpha_{\min} := \lambda_n(A(G))$. The corrected Case-B slot decomposition stated in Problem 7.1 features instead the *least*-magnitude G -negative eigenvalue $\alpha_{\text{top}} := \lambda_{n-n^-(H)}(A(G))$, which can be orders of magnitude smaller. For instance, on the L_{30} Case-B endpoint, $\alpha_{\min}^2 \approx 4.91$ but $\alpha_{\text{top}}^2 \approx 8.6 \times 10^{-4}$. Bounding $\alpha_{\min}^2$ from below does *not* bound α_{top}^2 from below, and the slot decomposition requires the latter. This $\alpha_{\min}$ -versus- α_{top} gap is the structural obstruction blocking any analytical $\delta^-(v^*) \geq 1$ claim; see Remark 5.15 and Problem 7.1.

Proposition 5.14 (Empirical b.minor census). *Across the b.minor working corpus of 2,235 Case-A and Case-B max-degsum records — covering enumerated 2-trees on $n \leq 10$ vertices, books B_k , the structured family $\text{BT}(k, 2)$, 2-paths L_n , and fans F_n — every max-degsum ear v^* satisfies*

$$\delta^-(v^*) \geq 1.29405\dots,$$

attained at the $n = 10$ Case-B record with graph6 `I}iSSGI@O` (vertex $v = 8$, family `enum`). In particular $\delta^-(v^) \geq 1$ holds on every record in the corpus. (The $n = 10$ caterpillar `I}qcHG`GO` cited in F10 of Section 6 is a different graph: it minimises the joint invariant $I(v)$ over the corpus, not $\delta^-(v^*)$.)*

Proof. Direct numerical verification: for each of the 2,235 records, the quantity $\delta^-(v^*) = s^-(G) - s^-(H)$ is computed in IEEE double precision via `numpy.linalg.eigvalsh` and checked against the threshold. The corpus minimum is recorded in `data/case_AB_census.json`, regression-locked by `tests/test_b_minor.py`. $\square$

Remark 5.15 (Status of the analytical $\delta^-(v^*) \geq 1$ claim). The analytical claim “ $\delta^-(v^*) \geq 1$ unconditionally on Case-B max-degsum ears” is *not* a theorem. An intermediate argument derived $\delta^- \geq \alpha_{\min}^2$ in Case B via a chordal-graph Sylvester refinement, but that derivation contained a sign error in the slot decomposition (recorded as F7 in Section 6.7). The corrected Case-B identity is

$$\delta^-(v) = \alpha_{\text{top}}^2 + \sum_{j \in J^-(H)} (\lambda_{j+1}^2 - \mu_j^2), \quad \text{each summand} \geq 0,$$

where α_{top} is the least-magnitude G -negative eigenvalue, *not* $\alpha_{\min}$. Hence Lemma 5.12 bounds the wrong eigenvalue for the slot decomposition (Remark 5.13), and the analytical gap to $\delta^-(v^*) \geq 17/16$ is unified across Cases A and B as the slot-shift sum lower bound Problem 7.1. Proposition 5.14 is the strongest empirically-supported statement currently available; no analytical lower bound of the form “ $\delta^-(v^*) \geq c$ ” for any explicit $c > 0$ on Case-B max-degsum ears is currently known.

5.8. Closing condition (a) on 2-paths via Stieltjes.

Theorem 5.16 ($I_\infty(L)$ closed form).

$$I_\infty(L) := \lim_{n \rightarrow \infty} I(L_n, v^*) = \frac{2(310\pi^2 - 837\sqrt{3}\pi + 2187)}{27\pi(20\pi - 27\sqrt{3})} \approx 1.0157.$$

In particular $I_\infty(L) > T = 0.4122$, so condition (a) of Conjecture 5.5 holds on the 2-path family in the asymptotic limit.

Proof. By Theorem 4.10, the negative-side spectral measure ν_∞^- of the limit operator at $w = e_1 + e_2$ is absolutely continuous on $(-9/4, 0)$ with density $\rho_w(\lambda) = \sin(\theta_2(\lambda) - \theta_1(\lambda))/\pi$, where $\theta_1, \theta_2 \in (\pi/3, \pi)$ are the two preimages of λ under $f(\theta) = 2\cos\theta + 2\cos 2\theta$. The substitution $\lambda = f(\theta_1)$ with

$\theta_2 = \arccos(-1/2 - \cos \theta_1)$ converts each moment integral into a one-variable trigonometric integral over $\theta_1 \in (\pi/3, \theta_{\min})$ with $\theta_{\min} = \arccos(-1/4)$:

$$W_{\infty}^{-}(L) = \int_{-9/4}^0 \rho_w d\lambda = 1 - \frac{3\sqrt{3}}{4\pi}, \quad M_{1,\infty}^{-}(L) = \frac{2}{3} - \frac{9\sqrt{3}}{4\pi}, \quad M_{2,\infty}^{-}(L) = 3 - \frac{81\sqrt{3}}{20\pi},$$

via direct integration; the closed forms are verified symbolically. Substituting into the candidate $I_{\infty}(L) = W_{\infty}^{-} + (M_{1,\infty}^{-})^2/M_{2,\infty}^{-}$ and simplifying gives the stated formula. The convergence $I(L_n, v^*) \rightarrow I_{\infty}(L)$ uses strong-resolvent convergence of $A(L_{n-1})$ to the half-line operator and a Portmanteau argument on the signed moments: $\lambda^k \mathbf{1}[\lambda < 0]$ is continuous at 0 for $k = 1, 2$, and the $k = 0$ moment uses the no-atom property of ν_{∞}^{-} at $\lambda = 0$.

Numerically $I_{\infty}(L) \approx 1.0157374\dots > 0.4122 = T$, with explicit slack ≈ 0.6035 . $\square$

Remark 5.17 (Residual open subobligations on the 2-path closure). Three residual subobligations of Theorem 5.16 are identified but do not affect the displayed limit value or the inequality $I_{\infty}(L) > T$. *Rate (O13.1)*: the convergence $I(L_n, v^*) \rightarrow I_{\infty}(L)$ is established qualitatively; an explicit n_0 from which $|I(v) - I_{\infty}(L)| \leq \varepsilon$ for given ε requires a non-simple-loop BBG analogue (cf. F9, Remark 4.5). *Branch (O13.2)*: the cube-root branch choices for the two roots ξ_1, ξ_2 of the characteristic quartic used in Theorem 4.10 are verified numerically at one complex base point; a formal analytic-continuation argument is light but not formalised. *Trig (O13.3)*: the trigonometric simplification in the proof above factors the intermediate expression $\Im(s^2 + s - p) + \sin(\theta_2 - \theta_1)$ via the half-angle product $2 \cos(\frac{\theta_1 + \theta_2}{2}) \cos(\frac{\theta_2 - \theta_1}{2}) = \cos \theta_1 + \cos \theta_2 = -1/2$, which writes the bracket as $2 \sin(\frac{\theta_2 - \theta_1}{2}) \cos(\frac{\theta_1 + \theta_2}{2}) [4 \cos(\frac{\theta_2 - \theta_1}{2}) \cos(\frac{\theta_1 + \theta_2}{2}) + 1] = 0$ on the spectral support; writing this out adds one short line of algebra. None of O13.1–O13.3 affects the displayed limit value or the inequality $I_{\infty}(L) > T$.

5.9. Status of the joint-invariant ansatz. We summarise the status of the two conditions of Conjecture 5.5 after the work of this section.

Condition (a) (structural). Proved on two subfamilies plus the common-spine-edge characterisation of books:

- Books B_k : Corollary 4.2, restated in clique-tree language as Proposition 5.10 with explicit value $I(v) = 2 - 2/\sqrt{8(n-2)-7}$.
- 2-paths L_n in the asymptotic limit: Theorem 5.16 with $I_{\infty}(L) \approx 1.0157$.

On the $\text{BT}(k, 2)$ family the max-degsum selector picks a clean book-page ear, and condition (a) is observed to hold empirically on the working corpus; an analytic closure is not given here, since the tail-ear analysis of Proposition 4.6 addresses only the universal form and does not by itself certify condition (a) at the selector ear uniformly in k . The two-sided Cauchy–Schwarz structural lower bound (Lemma 5.8) gives a corpus-uniform value of ≈ 0.396 , which is ≈ 0.016 short of $T = 0.4122$ at the worst case. Condition (a) on general

2-trees remains open (Problem 7.2); the residual obstruction is a *structural-sufficiency gap*, structurally distinct from the slot-shift wall of condition (b). Condition (b) (spectral). Two partial results: the conditional analytical bound $\alpha_{\min}^2 \geq 1$ at Case-B max-degsum ears (Lemma 5.12, under the sufficient moment condition $|M_1^-(v^*)| \geq W^-(v^*)(1 - W^-(v^*))$) and the empirical census $\delta^-(v^*) \geq 1.2941$ across 2,235 records (Proposition 5.14). Crucially, *no* analytical $\delta^-(v^*) \geq c$ lower bound is established for any explicit $c > 0$ on Case-B max-degsum ears (Remark 5.15); the bound Lemma 5.12 controls $\alpha_{\min}^2$, not δ^- , and the corrected slot decomposition involves α_{top} in place of $\alpha_{\min}$ (Remark 5.13). The headline $\delta^-(v^*) \geq 17/16$ form remains open on every subfamily, with the slot-shift sum bound Problem 7.1 the unified obstruction across Cases A and B.

6. FAILURE MODES

This section catalogues ten specific dead ends rejected during the search programme. Each failure mode is paired with a regression fixture in `tests/fixtures/` (or, when no falsifying graph is needed, a combinatorial witness) so the rejection is reproducible. We include this material in the main body (rather than an appendix) because the *retirement* of the universal ear lemma (Corollary 4.7) and of single-scalar candidates is as informative for future work as the theorems that survived.

6.1. F1. The universal ear-deletion lemma. The naive statement “every simplicial degree-2 ear v of every 2-tree on $n \geq 4$ vertices satisfies $\delta^-(v) \geq 17/16$ ” is false. Two witnesses: the asymptotic $\delta_{\infty}^-(\text{BT}) \approx 1.0353 < 17/16$ on the $\text{BT}(k, 2)$ tail family (Proposition 4.6; at $k = 50$ the value is already $\delta^- \approx 1.0575$), and a seed-149 random 2-tree on $n = 100$ vertices with $\delta^- \approx 1.0610$. The latter graph is preserved as the permanent regression fixture `tests/fixtures/two_tree_universal_counterexamples.json`; the former is the structured proof of Corollary 4.7. The correct reformulation is the *existential* ear-selection lemma (L') (Conjecture 3.1).

6.2. F2. The $\|w\|^2 = 4$ normalisation bug. The supporting-edge vector $w = e_a + e_b$ has $\|w\|^2 = 2$, not 4: the two standard basis vectors are orthogonal, $e_a^\top e_b = 0$, regardless of whether $\{a, b\}$ is an edge of H . The mistaken value 4 was conflated with the trace identity $\delta^+(v) + \delta^-(v) = 2 \deg_G(v) = 4$, which is a different (and correct) derivation ($\text{tr}(A(G)^2) - \text{tr}(A(H)^2) = 2 \deg_G(v)$). Consequences of the corrected normalisation $\sum_i c_i^2 = \|w\|^2 = 2$: the walk-moment identity Lemma 5.1 takes its stated form, the Cauchy–Schwarz pair Lemmas 5.2 and 5.8 have the stated factors, and the threshold $T = 0.4122$ is no longer derivable from a “natural” rescaling of $17/16$. Regression fixture: `tests/fixtures/w_norm_squared_is_2.json`.

6.3. F3. Single-scalar selector thresholds. The natural single-scalar thresholds — $W^-(v^*) \geq 17/16$ (the $\|w\|^2 = 4$ era) and $W^-(v^*) \geq 17/32$

(after the v9 $\|w\|^2 = 2$ renormalisation) — are both empirically false. Witnesses: L_5 endpoint has $W^- \approx 0.515$, L_6 has $W^- \approx 0.380$, and L_{12} has $W^- \approx 0.502$; all are below $17/32 = 0.53125$. An empirically-fitted threshold $T < 17/32$ does survive on a finite corpus, but the worst-case bad ears from the $\text{BT}(k, 2)$ tail family have $W^- \rightarrow 0$ as $k \rightarrow \infty$ while the max-degsum-ear infimum of W^- also drops, narrowing the window asymptotically. The candidate $I(v) = W^- + (M_1^-)^2/M_2^-$ at $T = 0.4122$ (Definition 5.4) is a genuine joint invariant, not a single-scalar rebrand. Regression test: `tests/test_lprime_subfamilies.py`.

6.4. F4. The naive sine-basis half-line boundary density. The naive guess for the half-line boundary density of the pentadiagonal Toeplitz operator with symbol $f(\theta) = 2 \cos \theta + 2 \cos 2\theta$ at $w = e_1 + e_2$ is $(\sin \theta + \sin 2\theta)^2/\pi$, by extrapolation from the tridiagonal case where the boundary density at e_1 is $\sin^2 \theta/\pi$. This naive density fails the Plancherel-norm test by a factor of 100+: direct projection of w onto the eigenbasis of $A(L_{n-1})$ at $n = 200$ gives entries that differ from $(\sin \theta_j + \sin 2\theta_j)^2 \cdot 2/(n+1)$ by factors of order 10^2 . The correct density is the *signed angle gap*

$$\rho_w(\lambda) = \frac{1}{\pi} \sin(\theta_2(\lambda) - \theta_1(\lambda))$$

on the negative-spectrum interval $(-9/4, 0)$, where $\theta_1 < \theta_2$ are the two preimages of λ under f (Theorem 4.10). The naive form fails because the eigenvectors of pentadiagonal Toeplitz are *not* simple sinusoids — a tridiagonal-specific structure that does not survive bandwidth 2.

6.5. F5. The asymmetric “selector + dual” framework. A natural reflex after the universal-lemma failure (F1) is to combine the max-degsum selector for δ^- with a dual selector for δ^+ , hoping the asymmetry between s^+ and s^- (Elphick–Linz [11]) yields a tighter ceiling on the positive side. Direct computational test on the same corpus rules this out: positive-side and negative-side Rayleigh-tightness ratios for Lemma B1 (and its positive-side dual) agree to within 3%, with no asymmetry to exploit. Moreover, the trace identity $\delta^+(v) + \delta^-(v) = 4$ at any degree-2 ear forces a one-to-one mapping between the two gains, so any one-sided selector for δ^- is automatically a selector for δ^+ at the reflected threshold; the dual framework collapses to a relabelling.

6.6. F6. The $\alpha\omega$ fallback route. Theorem 8.1 of [2] proves Conjecture 1.1 strictly under the hypothesis $\alpha(G)\omega(G) \leq n/17$. This route is uniformly inapplicable to the 2-tree class: every 2-tree has $\omega(G) = 3$ (treewidth 2 caps the max clique at a triangle), and the independence number $\alpha(G)$ scales linearly with n on every binding family — empirically on the corpus $(\alpha\omega)/n$ ranges between 0.8 (books) and 1.0 (paths), *never* below $1/17 \approx 0.059$. The constant $1/17$ in Theorem 8.1 traces back to the slack $\epsilon = 1/16$ of the P_3 -removal lemma; improving 17 requires sharpening that slack, which is

research upstream of our workstream. The $\alpha\omega$ -route is therefore not the right fallback for 2-trees.

6.7. F7. Sign error in Lemma B1 optimisation. An intermediate draft of Lemma 5.6 had the wrong sign in the Rayleigh-quotient stationary equation ($\beta^2 + (M_1^-/W^-)\beta - W^- = 0$ instead of the correct $\beta^2 - (M_1^-/W^-)\beta - W^- = 0$). This propagated into the identification of β_+ as the minimiser: the wrong-sign analysis identifies $\beta_+ = (|M_1^-| + \sqrt{(M_1^-)^2 + 4(W^-)^3})/(2W^-)$, whereas the correct minimiser is $\beta_+ = (\sqrt{(M_1^-)^2 + 4(W^-)^3} - |M_1^-|)/(2W^-)$. The displayed lemma bound happens to be unchanged (a luck-of-the-algebra coincidence: $R(\beta_+) = -W^-/\beta_+$ rationalises to the same final formula in both cases), but the intermediate Case-B slot decomposition built on the wrong root identification gave a false $\delta^-(v^*) \geq 1$ claim. Spot-check at L_5 catches it: $\lambda_{\min}(A(L_5)) = -1.618$ exactly, while the wrong-sign formula predicts $+1.387$. The corrected proof in Section 5.4 uses the right root throughout, and the falsified $\delta^-(v^*) \geq 1$ claim is downgraded to the conditional Lemma 5.12 plus the empirical Proposition 5.14.

6.8. F8. “Floating-point certified” $\neq$ interval-arithmetic certified. The $\delta^-(L_n) \geq 17/16 + 1/4$ certificate of Theorem 4.8 uses `numpy.linalg.eigvalsh` with an *a-posteriori* Demmel–Kahan forward-error bound [9, Thm. 5.5]. The resulting statement is engineering-rigorous *conditional on* the `dsyevd` envelope $c \leq 10$ used in `scripts/mpmath_certify.py` (the original analysis used the deliberately looser working envelope $c \leq 10^4$, under which the certificate is *a fortiori* also valid): under the looser $c \leq 10^4$ envelope the forward error on each computed eigenvalue is provably bounded by $8.88 \cdot 10^{-9}$ at $n \leq 1000$ and the propagated δ^- forward error by $1.42 \cdot 10^{-4}$, far below the slack 0.257; the envelope itself is plausible (over an order of magnitude above any published constant for backward-stable symmetric eigensolvers) but not formally derived for the exact LAPACK `dsyevd` pipeline within this paper, and the certificate is *not* interval arithmetic. A truly interval-arithmetic certificate (via `mpmath` at, say, 50 decimal digits, or via Krawczyk-style verified eigensolver) would be a strictly stronger guarantee. For our purposes the *a-posteriori* bound suffices; the upgrade path is mechanical.

6.9. F9. Simple-loop hypothesis for BBG-type asymptotic constants. Bogoya–Böttcher–Grudsky [7] and the related Avram–Parter / Widom-type effective constants for finite- n corrections to Szegő’s theorem assume a *simple-loop* symbol: f traces the unit circle once as θ traverses $[-\pi, \pi]$, without singularities. Our pentadiagonal symbol $f(\theta) = 2 \cos \theta + 2 \cos 2\theta$ has zeros at both $\theta = \pi/3$ (interior, transversal) and $\theta = \pi$ (boundary), so the simple-loop hypothesis fails. The *rate* of convergence in Theorem 4.10 — i.e. an explicit n_0 from which $|I(v) - I_\infty(L)| \leq \epsilon$ for given ϵ — is therefore not available by off-the-shelf citation. The limit value $I_\infty(L) \approx 1.0157$ and its closed

form are unaffected; the rate is recorded as open subobligation O13.1 of the workstream.

6.10. F10. The 2-path is not the universal binding case for $I(v)$. An intermediate working hypothesis held that the 2-path family asymptotic $I_\infty(L) \approx 1.0157$ is the binding case for the joint invariant $I(v)$ at the max-degsum ear of every 2-tree, i.e. that $I(v^*) \geq I_\infty(L)$ would close condition (a) on general 2-trees. This is false at finite n : at $n = 10$ the caterpillar 2-tree with graph6 `I}qcHG`G0` satisfies $I(v^*) = 0.6384$, which is below $I_\infty(L)$ by ≈ 0.38 and below $I(v)$ on L_n for every $n \in \{4, \dots, 30\}$.

Two consequences. First, the open problem Problem 7.2 is correctly framed as $I(v^*) \geq T = 0.4122$, not as $I(v^*) \geq I_\infty(L)$; the corpus minimum gives slack ≈ 0.23 above T at the worst case. Second, a 2-path-embedding monotonicity argument cannot close condition (a) on general 2-trees, since the caterpillar minimiser embeds a 2-path that has $I(v) > 0.6384$ on its own (the 2-path on 5 vertices already has $I(v) = 0.79\dots$). The structural lower bound Lemma 5.8 — a moment-form functional rather than a subfamily comparison — is therefore the right tool. The caterpillar minimiser is preserved as a permanent regression fixture in `tests/fixtures/joint_invariant_falsified.json`.

6.11. Summary table. Appendix B contains a tabular summary of failure modes F1–F10, with columns: identifier, the precise false statement, the smallest falsifying witness (combinatorial or numerical), and the regression fixture file.

7. OPEN PROBLEMS AND OUTLOOK

We have proved (L') on two subfamilies (books and the 2-path asymptotic), exhibited the bad-tail-ear asymptotic on $\text{BT}(k, 2)$ that refutes the universal form, partially closed condition (b) of the joint-invariant ansatz on Case-B max-degsum ears, and produced a two-sided Cauchy–Schwarz structural lower bound (Lemma 5.8) on $I(v)$ that falls ≈ 0.016 short of T at the worst empirical case. Empirically, the minimum of $I(v^*)$ across the working corpus is 0.6384, attained at an $n = 10$ caterpillar 2-tree (graph6 `I}qcHG`G0`); the 2-path is *not* the universal binding case (F10). Two critical-path items remain: condition (b) on general 2-trees (Problem 7.1, the slot-shift wall) and condition (a) on general 2-trees (Problem 7.2, a structural-sufficiency gap).

7.1. Problem A: condition (b), the slot-shift wall.

Problem 7.1 (The slot-shift wall). Let G be a 2-tree on $n \geq 4$ vertices, v a simplicial degree-2 ear of G with $I(v) \geq T$, and write $J^-(H) := \{j : \mu_j(H) < 0\}$. Establish either of the following two cases.

- *Case A* ($n^-(G) = n^-(H)$):

$$\sum_{j \in J^-(H)} (\lambda_{j+1}(G)^2 - \mu_j(H)^2) \geq \frac{17}{16}.$$

- *Case B* ($n^-(G) = n^-(H) + 1$):

$$\alpha_{\text{top}}^2 + \sum_{j \in J^-(H)} (\lambda_{j+1}(G)^2 - \mu_j(H)^2) \geq \frac{17}{16},$$

where $\alpha_{\text{top}} := \lambda_{n-n^-(H)}(G)$ is the *least*-magnitude G -negative eigenvalue.

The slot-shift sum is the unified bottleneck across Cases A and B; in Case B the extra non-negative term α_{top}^2 helps nominally but can be as small as $\sim 10^{-3}$ in absolute value (Remark 5.13), so it does not by itself supply the slack. Standard perturbation tools (Lehmann–Goerisch, Temple, Aronszajn) are too weak in the cancellation regime where α_{min}^2 and μ_{n-1}^2 are both large while their difference is small. The BT(50, 2) tail-ear example has $\alpha_{\text{min}}^2 \approx 90$ and $\mu_{n-1}^2 \approx 90$ but $\delta^- \approx 1.06$, illustrating the cancellation. Plausibly the right tool is a quantitative strict-slack refinement of the chordal-graph Sylvester argument that gives the trivial floor $\delta^-(v) \geq 0$, or a Cauchy-style integral representation of the slot-shift sum via residues of the secular function $\lambda \mapsto \sum_i c_i^2 / (\lambda - \mu_i)$.

7.2. Problem B: condition (a) on general 2-trees.

Problem 7.2 (Condition (a) on general 2-trees). Show that $I(v^*) \geq T = 0.4122$ for every 2-tree on $n \geq 4$ vertices, where v^* is the max-degsum simplicial degree-2 ear.

The corpus minimum of $I(v^*)$ is 0.6384, attained at an $n = 10$ caterpillar 2-tree and *not* on the 2-path family (F10), so empirically there is ≈ 0.23 of slack above T at the worst case. The two-sided Cauchy–Schwarz structural lower bound Lemma 5.8 achieves ≈ 0.396 uniformly on the corpus — ≈ 0.016 below T — so the existing structural toolkit is *almost* sufficient but not quite. The residual obstacle is a *structural-sufficiency gap*: the four moments $(W^-, W^0, |M_1^-|, M_2)$ at a simplicial ear are insufficient to pin $I(v)$ at finite n on the caterpillar minimiser. A natural sharpening of Lemma 5.8 incorporates a fifth clique-tree quantity capturing the position of the ear’s leaf triangle within $T(H)$ — for example the diameter or eccentricity of the leaf in the clique tree, or a depth invariant — and would yield a five-variable lower bound. A clique-tree Schur-complement induction with this added bookkeeping is the natural attack. The residual obstacle is *structurally distinct* from the slot-shift wall of Problem 7.1. The reduction map of Section 3.6 is a conjunction of conditions: under condition (a) (Item (a)) the max-degsum ear clears $I(v^*) \geq T$, and under condition (b) (Item (b)) that threshold lifts to $\delta^-(v^*) \in [17/16, 47/16]$. Hence

$$(L') \text{ at } v^* \iff (\text{condition (a)}) \wedge (\text{condition (b)}).$$

Closing Conjecture 1.1 on 2-trees through the infrastructure of this paper requires closing *both* Problem 7.2 (condition (a)) *and* Problem 7.1 (condition (b)) — or supplying an independent bridge that bypasses one of the two conditions. Closing either problem alone, in isolation, does not suffice.

7.3. Outlook. Three further directions are worth flagging.

First, on the 2-tree front: the $\text{BT}(k, t)$ family for $t \geq 3$ is the natural next adversarial input. The universal lemma fails worse there than at $t = 2$ — the asymptotic bad-ear δ^- drops below 1.0353 as t grows — but the selector lemma is empirically robust; verifying that condition (a) of Conjecture 5.5 extends uniformly to $\text{BT}(k, t)$ would sharpen Proposition 5.14 into an honest pre-asymptotic statement on a two-parameter family.

Second, beyond 2-trees: the residue-control programme of [2, §8] and the chordal-graph generalisation of our clique-tree machinery (Lemmas 5.1 and 5.3) are natural next targets. The walk-moment identity $M_2 = \sigma(v) + 2|T_{ab}(H)|$ is specific to graphs where the supporting edge lies in triangles, but generalises to chordal graphs of treewidth k with appropriate k -clique counting.

Third, Conjecture 1.1 in full remains open and was assessed by the authors of [2] as one of the central open problems in the area. Closing it on 2-trees is a tractable slice, not a known step in an induction toward the full conjecture; the analytical content required for general chordal graphs, let alone arbitrary graphs, is plausibly orthogonal to the moment-form framework developed here.

APPENDIX A. REPRODUCIBILITY AND SOFTWARE SUPPLEMENT

A.1. Repository pointer. A software supplement accompanies this paper at https://github.com/<TBD>/positive_square_energy_equality, git commit hash TBD. The supplement contains:

- the corpus of 2-tree adjacency data (`data/two_trees_n*.json`, `data/joint_invariant_scan*.json`, `data/case_AB_census.json`);
- all scripts used in the numerical proofs (`scripts/`), including `spectrum_check.py`, `two_tree_enum.py`, `mpmath_certify.py`, `joint_invariant_features.py`, `half_line_stieltjes.py`;
- the test suite (`tests/`), 539 tests passing at submission, regression-locking each theorem and failure mode to a named test or fixture file;
- the companion technical notes referenced throughout the paper (`docs/lprime_*.md`).

The supplement is licensed under the BSD-3 clause; all data files are in human-readable JSON or graph6 plain-text.

A.2. The working corpus. The numerical claims of the paper rest on the following corpus of 2-trees:

- Enumerated 2-trees with $n \in \{4, \dots, 10\}$: 724 graphs (enumerated up to isomorphism by the standard 2-tree construction `scripts/two_tree_enum.py`).
- Random 2-trees with $n \in \{15, 20, 30, 50, 100\}$, 50 seeds each: 250 graphs (constructed by attaching each new vertex to a uniformly random existing edge, with reproducible seeding via `random.Random(seed)`).

- $\text{BT}(k, 2)$ for $k \in \{2, 5, 10, 25, 50, 100\}$: 6 structured graphs.
- Books B_k for $k \in \{2, \dots, 30\}$: 29 graphs.
- 2-paths L_n for $n \in \{4, \dots, 30\}$: 27 graphs.
- Fans F_n for $n \in \{4, \dots, 30\}$: 27 graphs.

Total: 1063 graphs, 2628 max-degsum ear records, 8890 all-ear records. The b.minor census of Proposition 5.14 uses the 2235-record subset that excludes the random-2-tree contribution (restricting to enumerated, books, BT, L_n , F_n).

A.3. The Demmel–Kahan certificate. The precision protocol used to certify Theorem 4.8:

- (1) Compute eigenvalues $\tilde{\lambda}_i(A(L_n))$ for each $n \in [4, 1000]$ in IEEE-754 double precision via `numpy.linalg.eigvalsh`, whose default backend is LAPACK’s `dsyevd` (Householder tridiagonalisation followed by symmetric divide-and-conquer).
- (2) Apply the Demmel–Kahan-style a-posteriori bound [9, Thm. 5.5]: the forward error on each eigenvalue satisfies $|\tilde{\lambda}_i - \lambda_i| \leq C_n \cdot \|A(L_n)\|_2 \cdot \varepsilon_{\text{mach}}$ with $C_n \leq cn$ for an absolute constant c from the backward-error analysis. We take the conservative working envelope $c \leq 10^4$ (overshooting published constants for backward-stable symmetric eigensolvers by at least one order of magnitude; see Remark 4.9 for the honest status of this constant — it is consistent with the literature on the Householder-plus-divide-and-conquer pipeline used by `dsyevd`, but a fully formal derivation of an explicit `dsyevd`-specific envelope constant is beyond the scope of this paper). With $\|A(L_n)\|_2 \leq \|f\|_\infty = 4$ the per-eigenvalue bound is $\leq 8.88 \cdot 10^{-12} n \leq 8.88 \cdot 10^{-9}$ at $n \leq 1000$, and the propagated $\tilde{s}_n^-$ error is $\leq 7.1 \cdot 10^{-5}$ uniformly across $n \in [4, 1000]$ (cf. Theorem 4.8).
- (3) The numerical slack is $\tilde{\delta}_n^- - 17/16 \geq 0.257$ for every $n \in [4, 1000]$, with minimum attained at $n = 6$. After subtracting the propagated δ^- forward-error bound $\leq 1.42 \cdot 10^{-4}$, the certified slack is $\geq 0.257 - 1.5 \cdot 10^{-4} > 1/4$.

The certificate is reproduced end-to-end by `tests/test_mpmath_certify.py`.

The stored mpmath companion certificate at `data/two_path_mpmath_certificate.json` consists of 17 records (a curated subset $n \in \{4, 5, \dots, 10, 12, 15, 20, 30, 50, 80, 100, 130, 160, 200\}$ chosen to include the empirical worst-case $n = 6$ together with a logarithmically-spaced tail) computed at 50 decimal digits via `mpmath.eigsys`; for runtime budget the test suite spot-checks 9 of these cases ($n \in \{4, 5, 6, 7, 8, 10, 20, 50, 100\}$) at 30 decimal digits, which still gives ~ 25 digits of headroom on every computed quantity. The full 50-digit records are present in the JSON for offline inspection. All checks pass.

A.4. The Stieltjes-transform identification. The closed-form value $I_\infty(L) = 2(310\pi^2 - 837\sqrt{3}\pi + 2187)/(27\pi(20\pi - 27\sqrt{3}))$ of Theorem 5.16 is derived as follows.

- (1) *Boundary density.* On the negative branch $\lambda \in (-9/4, 0)$ the half-line spectral density of $A(L_\infty)$ tested against $w = e_1 + e_2$ is the signed angle gap $\rho_w(\lambda) = \sin(\theta_2 - \theta_1)/\pi$ (Step 3 of the proof of Theorem 4.10).
- (2) *Moments.* Substituting $\lambda = f(\theta_1)$ with $\theta_2 = \arccos(-1/2 - \cos \theta_1)$, change of variables and trigonometric integration give

$$W_\infty^- = 1 - \frac{3\sqrt{3}}{4\pi}, \quad M_{1,\infty}^- = \frac{2}{3} - \frac{9\sqrt{3}}{4\pi}, \quad M_{2,\infty}^- = 3 - \frac{81\sqrt{3}}{20\pi}.$$

The script `scripts/half_line_stieltjes.py` performs the sympy computation symbolically and verifies against numerical quadrature at 20-digit precision.

- (3) *Closed-form assembly.* $I_\infty(L) = W_\infty^- + (M_{1,\infty}^-)^2/M_{2,\infty}^-$ simplifies to the displayed rational function of π and $\sqrt{3}$. Numerically $I_\infty(L) \approx 1.01574$, exceeding $T = 0.4122$ by ≈ 0.604 .

Reproducibility: the symbolic $I_\infty(L)$ closed form, the moment expressions, the $G_w = s^2 + s - p$ collapse, and the spectral-density formula $\rho_w = \sin(\theta_2 - \theta_1)/\pi$ are regression-locked at the sympy level by `tests/test_half_line_stieltjes.py` (in particular, `test_symbolic_moments_match_candidate_closed_form` and `test_I_inf_closed_form` verify the symbolic differences are exactly zero, and `test_unsigned_moments_match_matrix_moments` crosschecks the full-spectrum unsigned moments against $(M_0, M_1, M_2) = (2, 2, 7)$ by quadrature). The finite- n convergence harness for $\delta^-(L_n) \rightarrow \delta_\infty^-$ on $n \in [4, 200]$ is in `tests/test_two_path_widom_tightness.py`.

A.5. Theorem-to-test cross reference. Each numbered statement of the paper is regression-locked to a named test file:

- Theorem 4.1 $\rightarrow$ `tests/test_lprime_subfamilies.py`
- Theorem 4.3 $\rightarrow$ `tests/test_lprime_subfamilies.py`
- Proposition 4.6 $\rightarrow$ `tests/test_lprime_subfamilies.py`
- Theorem 4.8 $\rightarrow$ `tests/test_mpmath_certify.py`
- Theorem 4.10 $\rightarrow$ `tests/test_half_line_stieltjes.py` (symbolic closed forms for ρ_w , the boundary system, and $G_w = s^2 + s - p$); `tests/test_two_path_widom_tightness.py` (finite- n convergence harness)
- Lemma 5.1 $\rightarrow$ `tests/test_w_norm_squared_invariant.py`
- Lemma 5.6 $\rightarrow$ `tests/test_joint_invariant_candidates.py`
- Lemma 5.8 $\rightarrow$ `tests/test_lprime_5e_a_general.py`
- Proposition 5.10 $\rightarrow$ `tests/test_lprime_subfamilies.py`
- Lemma 5.12, Proposition 5.14 $\rightarrow$ `tests/test_b_minor.py`, `tests/test_max_degsum_selector.py`
- Theorem 5.16 $\rightarrow$ `tests/test_half_line_stieltjes.py` (symbolic verification of the $I_\infty(L)$ closed form, including the symbolic equality of the three negative-branch moments and of the assembled $I_\infty(L)$ expression); `tests/test_two_path_widom_tightness.py` (the $n \leq 200$ finite- n convergence harness, including the trace identity and $O(1/n)$ tail constant)

Failure modes F1–F10 of Section 6 are each backed by either a regression fixture in `tests/fixtures/` or an analytical witness; see Appendix B for the table.

APPENDIX B. FAILURE-MODE FIXTURES

Tabulated witnesses for the failure modes F1–F10 of Section 6. Each row records: the identifier; the precise false statement (in one line); the smallest witnessing graph (described combinatorially or by graph6 string, with the relevant numerical quantity); and the regression fixture file in the repository (or an em-dash if the witness is analytical / structural).

ID	False statement	Smallest witness	Fixture
F1	Every simplicial degree-2 ear of every 2-tree satisfies $\delta^- \geq 17/16$.	BT($k, 2$) tail at $k \geq 40$ (direct computation: $\delta_{\text{tail}}^-(\text{BT}(40, 2)) \approx 1.0623 < 1.0625 = 17/16$, with asymptote $4 - \alpha^2 + \beta^2 \approx 1.0353$, Proposition 4.6); also a random 2-tree on $n = 100$ (seed 149).	two_tree_universal_counterexamples.j
F2	$\ w\ ^2 = 4$.	Any 2-tree ear (algebraic).	w_norm_squared_is_2.json
F3	$W^-(v^*) \geq 17/32$ (or $17/16$).	L_5 endpoint, $W^- = 0.515$.	joint_invariant_falsified.json
F4	Half-line boundary density is $(\sin \theta + \sin 2\theta)^2 / \pi$.	Pentadiagonal $f, \theta = \pi/3$.	(analytical; numerical mismatch factor 100-
F5	Asymmetric “selector + dual” for δ^+ .	Symmetric trace identity $\delta^+ + \delta^- = 4$.	— (no separate dual exists)
F6	$\alpha\omega \leq n/17$ regime covers 2-trees uniformly.	L_{30} vs B_{15} .	—
F7	Lemma B1 minimiser at β_- .	L_5 spot check ($\lambda_{\min} = -1.618$ vs predicted $+1.387$).	—
F8	“Floating-point certified” is rigorous without <i>a-posteriori</i> bound.	—	—
F9	Standard BBG / Avram-Parter constants apply to $f = 2 \cos + 2 \cos 2$.	Boundary zero at $\theta = \pi$ violates simple-loop hypothesis.	—
F10	The 2-path is the universal binding case for $I(v)$ at the max-degsum ear, i.e. $I(v^*) \geq I_\infty(L) \approx 1.0157$ on every 2-tree.	$n = 10$ caterpillar (graph6 I}qCHG`G0), $I(v^*) = 0.6384 < I_\infty(L)$.	joint_invariant_falsified.json

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